



Dual polarized antenna with high directivity for base station antennas

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Abstract

A new approach for designing antennas with high directivity and dual polarization, is investigated in the following thesis work.

The Yagi-Uda array with cross dipole elements and the multi-patch structures may be considered as possible solutions. The multi-patch structure is being investigated here for the following reasons: it is simple to build and provides a dual polarized radiation if properly excited, it is a relatively new concept which has been studied only experimentally. No theoretical insight has been given in the literature to explain the mechanism behind this antenna.

In order to study the multi-patch structure the full-wave approach has been chosen. This involves setting up a system of integral equations which are then solved using the moment method. This is known to give accurate solutions, although requiring a lot of computational power and computer memory.

By calculating the unknown equivalent currents on the patches we explain the coupling mechanism between them and the radiation mechanism behind this antenna. Apart from directivity, co-polarization and cross-polarization we investigate also the radiation bandwidth behavior. The impedance bandwidth is out of our scope. We will show that conclusions from the Yagi-Uda arrays can be used to explain its behavior, optimization hints from linear arrays provide useful information for the multi-patch structure and that this structure can also be regarded as supporting a travelling wave.

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Glossary

EFIE -Electric Field Integral Equation

EMF -Electromagnetic Field

F/B -Front to Back Ratio

GP -Ground Plane (Patch)

HPBW -Half Power BeamWidth

MFIE -Magnetic Field Integral Equation

MoM -Moment Method

PEC -Perfect Electric Conductor

TE -Transverse Electric

TM -Transverse Magnetic

UMTS -Universal Mobile Telecommunication System

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Chapter 1

Introduction

1.1 Problem statement

This thesis work is a theoretical study on dual polarized antennas with high directivity. We also present possible optimization hints.

Today, mobile communications networks have 3-sectors cells where dual polarized antennas with beamwidths of 60° to 90° are used. The growing capacity demand from the users as 3G and UMTS networks begin to be deployed may trigger the use of sites with more than 3 sectors. Here, antennas with more narrow beamwidths, close to 30° to 45° as the one investigated here may play a significant role.

In search of solutions for the above problem, the Yagi-Uda array with elements consisting in two dipoles in a cross configuration or the multiple patch structure antenna appear to be good candidates.

The Yagi-Uda array with cross elements is the natural solution that follows the initial single polarized antennas which had broadside arrays of dipoles. The cross element will provide a dual polarized beam. Questions to be asked here concern the separation between the two polarizations and difficulty of manufacturing. On the other hand, patch antennas are quite easy to build and can easily provide dual polarized radiation when properly feed. The directivity of these type of antennas is the main concern since usual beam-widths are around 60° to 120° .

One can easily see a possible parallel between the Yagi-Uda arrays and the multi-patch structures. Knowing the end-fire radiation condition of Yagi-Uda arrays, can we draw a similar condition for the multi-patch structure? Similarly, can we draw conclusions for the relations between patches size and spacing and radiation properties of this antenna? Is the extensive optimization work done so far on Yagi-Uda arrays useful in any sense for the multi-patch structure?

The solution investigated here is the multi-patch structure. We will show that a properly spaced and dimensioned patch structure can provide an end-fire radiation with the desired beamwidth and dual polarization capabilities. Also some conclusions on the radiation mechanism, end-fire radiation condition and some optimization hints for more narrow beamwidths and reduced back radiation will be given. We will show that the coupling mechanism does not depend on the feeding type, for example probe feed or aperture feed antennas.

Knowing this, the optimization conclusions will have a general value even if the far field patterns are different for the two type of feeding.

1.2 Previous Work

1.2.1 Yagi-Uda arrays:

One can classify antennas by looking at the phase distribution of the current along the antenna. There are antennas that exhibit constant current and voltage standing wave patterns formed by reflections from the ends. They are referred as standing wave or resonant antennas and an example is the dipole antenna. On the other hand, one can design antennas that have a uniform or non-uniform pattern in current and voltage, referred as travelling wave or nonresonant antennas. A progressive phase pattern is usually associated with the current and voltage distribution.

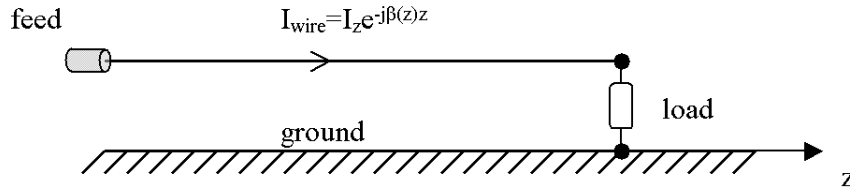


Figure 1.1: A travelling wave antenna - the long wire antenna. Phase pattern is a function of position, $\beta = \beta(z)$.

The Yagi-Uda antennas are the most common and known type of travelling wave antennas. Since their discovery in the 1920s they have received exhaustive analytical and experimental investigations in the literature. The simplicity of this antenna, together with its features, made it the proper choice for both amateur or professional antenna applications. This antenna has found applications from short waves to microwaves.

Yagi designed this antenna as a "wave canal". The excited dipole or the feed element has the role of initiating a surface travelling wave which is sustained by the conditions set on the directors and reflectors, meaning reflectors should have an inductive behavior and reflect the travelling wave while directors should have capacitive behavior and reinforce the EMF providing the end-fire radiation. A lot of experimental work is provided in the literature on finding the optimum spacing and length of the dipoles for maximum gain and front to back ratio [2].

The Yagi-Uda array can be described as a structure supporting a travelling wave. This surface travelling wave is described by the current distribution in each element and by the phase velocity of the travelling wave. This means that these arrays can act as a wave guide under certain conditions. A thorough theoretical investigation in this sense was done in [3] and [4] by R. J. Mailloux, on infinite, semi-infinite and long Yagi-Uda arrays. Arrays of uniformly spaced, identical elements were considered and a wave analysis was carried out using King and Sandler array theory, [5].

Some important conclusions need to be stated here:

- Only slow waves are allowed.
- Surface wave conditions.
- Optimization guide lines.

A necessary condition for the existence of a guided wave along the array is that there should not be radiation in directions transverse to the axis of propagation. This criterion is shown to require that $\beta_m^2 > k_0^2$, or that all waves be slow. β_m is the surface wave number, while k_0 is the free space wave number, [3].

Surface-wave theory has long been used to describe waves which are allowed to propagate along an interface, and the structure needed to support a given type of wave. For continuous one-dimensional and two-dimensional structures, a capacitive surface is needed to sustain a TE wave, an inductive surface for a TM wave. It has been found that no surface mode could be excited for dipoles of length $\lambda > l > 0.5\lambda$. Sengupta [6] using the transmission line analysis, showed that in some sense the antennas should have capacitive properties, and finally Serracchioli [7] obtained a numerical solution for the propagation constant β and found no solutions except for capacitive dipoles.

Figure 1.2 from [5] shows the region where a surface wave can exist. The considered case is the infinite Yagi-Uda array but conclusions are relevant also for finite structures. When the group velocity of the wave reaches the critical value, which can be regarded as a cut-off, there is no power flow in either direction, just energy storage. Beyond this point the array doesn't poses end-fire radiation at all, [5].

In [8] a correspondence between the phase diagram and the far field pattern was investigated and general suggestions were given. It was found that by decreasing the phase velocity, the actual directivity of the antenna was increased, although for excessive phase delays the radiation pattern was deteriorating in terms of back and side lobes.

Since the total field is obtained as:

$$E(total) = [elementfactor] \times [arrayfactor] \quad (1.1)$$

the directivity can be increased significantly by improving only the array factor(AF).

Figure 1.3 reproduced from [8], shows how phase delay is introduced by making the director of a 8 element Yagi-Uda array longer. As the phase delay is increased, the directivity of the antenna increases but up to a limit as stated above. For details see [8].

1.2.2 Patch antennas

The microstrip antenna, also known as patch antenna has received considerable attention since the 1970s. It's low profile, small size, cheap cost made it quite interesting for applications in civil and military communications. Some disadvantages are the low efficiency, high Q, poor polarization purity and normally very narrow frequency bandwidths.

This antenna usually consists in a radiating patch of different shapes (square, rectangle, dipole, disc, etc...) with or without dielectric substrate, which can be fed in different ways. The most common feeds are the microstrip feed line, the probe feed, the aperture coupled feed or the proximity-coupled feed [1].

Different methods of analysis have been developed for this antenna:

The transmission line model. This model gives good physical insight but is less accurate.

The cavity model offers more accurate results but is more complex.

The full-wave approach is the most accurate and versatile. It can also treat single elements or stacked patches of different shapes. These models include the integral equation formulation and the problem is solved using different numerical methods like: the method of moments, finite elements method, FDTD methods.

Since this work concerns stacked patches, the latter approach has been chosen.

Some results from the transmission line model and the cavity model are given in [1]. Since the current work concerns stacked patches we will focus next

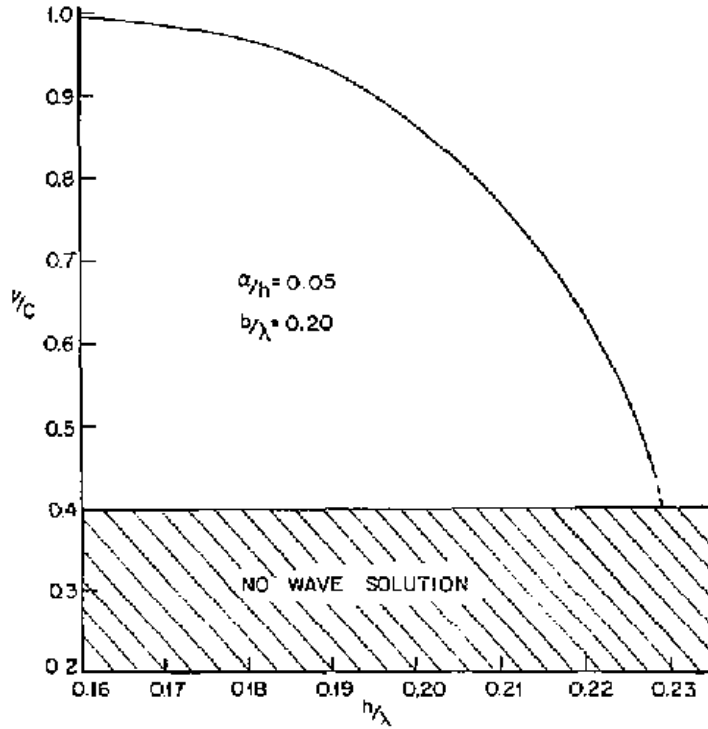


Figure 1.2: Velocity of propagation along infinite Yagi structures. Reproduced from [5]. a is the dipole's radius, b is the dipole spacing and $l = 2h$ is the dipole length.

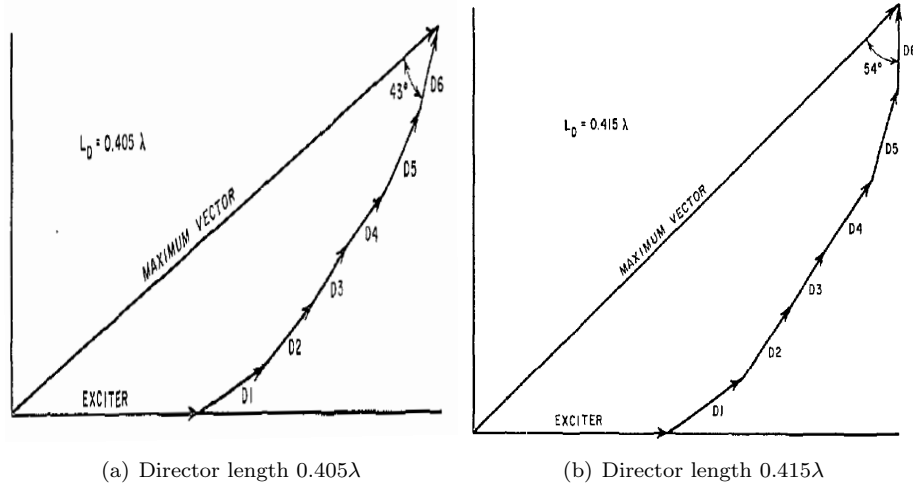


Figure 1.3: Phase diagram, 8 elements linear array, [8]

on some results found during the literature study on this subject. We should note from the beginning that many experimental suggestions have been found on how to choose the patch size and spacing in order to optimize the structure for directivity or bandwidth.

One example in this sense is the experimental study performed by Lee and Lee [9], on a two layer stacked patch antenna. Both patches are on dielectric substrates and the experiment is carried out using either air or teflon separation between them. Depending on the parasitic patch spacing three working regions are identified:

- A wide bandwidth region for spacing $< 0.15\lambda$, where bandwidths of $\approx 8\%$ to 18% were measured.
- An abnormal radiation region, where the pattern has some strange characteristics.
- A high gain region (8 to 11dB) for spacing $> 0.3\lambda$, and very narrow bandwidth ($< 2.5\%$)

There was also a noted influence on the patterns when the dielectric was air or teflon, teflon dielectric produced narrower beamwidth for both E and H planes.

An experimental work on back lobe reduction of patch antenna was carried out by Pozar, [10]. The reflector was used in combination with technics for increasing the bandwidth by widening the aperture. It provided the patch antenna with a F/B ratio of more than 20dB for the entire working bandwidth and by modifying the dimensions and the positions of it, the back null could be steered to different angles.

Finally, a three element stacked microstrip antenna was investigated by Egashira, [11]. Two parasitic elements of disc shape are placed above a round microstrip antenna and their influence is studied. The first element, i.e. the closest to the patch was denoted radiating element and it was used to increase

the overall gain, while the second one was denoted director and was used to increase the bandwidth. Gains of 12dBi and HPBW of $\approx 50^\circ$ were achieved for $\frac{\lambda}{4}$ director spacing.

As we can see from above, most of the results contain experimental data and provide some hints on how to dimension the stacked patches structure for different requirements. They don't provide though, any theoretical background or conclusion on the mechanism behind this structure. This is the main purpose of this work. By using the full-wave approach and modern numerical techniques (moment method) we will try to answer questions like: How do the patches couple together and what parameters influence this? Is the radiation always end-fire as desired, or is there a condition similar to the Yagi-Uda array? Also, some basic optimization rules or hints should be drawn to be used in building the actual antenna.

Although some tough theoretical questions must be answered, one should not forget the practical purpose pursued here. Possible applications of this antenna are multi-sector antennas for base stations, base-station antennas for cells along highways or railways with narrow beamwidth and repeater links.

Chapter 2

Theoretical background

2.1 General description of a scattering problem.

A typical radiation problem involves the computation of the radiated fields knowing the sources. In the case of an antenna, the sources are the currents on its elements. For the given problem, the sources are not specified, so a different approach must be taken.

This is a typical scattering problem which requires the setting up and solving a set of equations for the total fields in the surroundings of the scatterer by enforcing some specific boundary conditions. As this is not an easy task to do, one would simplify it and try to convert it into an equivalent problem for which a formal solution may be written. One way of doing this is to replace the inhomogeneous dielectric and magnetic material present in the problem by equivalent induced polarization currents and charges. The following derivations follow the material presented in [14]. One could write Maxwell's equations as follows:

$$\nabla \times \bar{E} = -j\omega\mu_0\bar{H} - \bar{M} \quad (2.1)$$

$$\nabla \times \bar{H} = j\omega\epsilon_0\bar{E} + \bar{J} \quad (2.2)$$

$$\nabla(\epsilon_0\bar{E}) = \rho_e \quad (2.3)$$

$$\nabla(\mu_0\bar{H}) = \rho_m \quad (2.4)$$

where now $\bar{J}$ and $\bar{M}$ are equivalent electric and magnetic sources radiating in free space. The new task of finding electromagnetic fields in free space is much straightforward than the original problem. So far these sources are unknowns to be determined.

The scatter in figure (2.1) is illuminated by a field produced by primary sources located somewhere outside the scatter. We have showed that the inhomogeneous material can be replaced by equivalent induced sources radiating in free space. Our task is to compute the total radiated field outside the scatterer volume, V_1 . The total field has a contribution from the primary sources in the absence of the scatter and another associated with the equivalent induced sources. We can denote the fields produced by the primary sources with: $\bar{E}^{inc}$, $\bar{H}^{inc}$ while the ones radiated by the equivalent sources, the scattered fields with

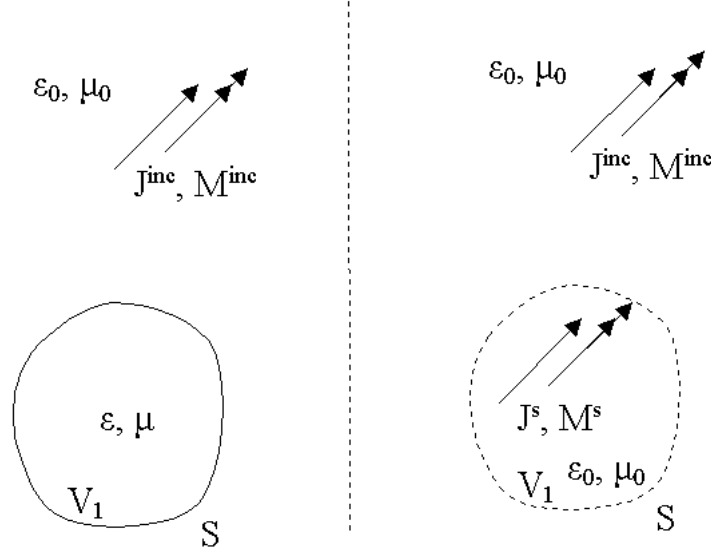


Figure 2.1: Scattering problem.

$\bar{E}^s$ and $\bar{H}^s$. The superposition of these fields will give the original fields in the presence of the scatterer:

$$\bar{E}^{tot} = \bar{E}^{inc} + \bar{E}^s \quad (2.5)$$

$$\bar{H}^{tot} = \bar{H}^{inc} + \bar{H}^s \quad (2.6)$$

It can be shown that the incident fields in the immediate vicinity of the scatterer satisfy the vector Helmholtz equations:

$$\nabla^2 \bar{E}^{inc} + k^2 \bar{E}^{inc} = 0 \quad (2.7)$$

$$\nabla^2 \bar{H}^{inc} + k^2 \bar{H}^{inc} = 0 \quad (2.8)$$

and the scattered fields are the solutions to the equations:

$$\nabla^2 \bar{E}^s + k^2 \bar{E}^s = j\omega\mu_0 \bar{J} - \frac{\nabla \nabla \cdot \bar{J}}{j\omega\epsilon_0} + \nabla \times \bar{M} \quad (2.9)$$

$$\nabla^2 \bar{H}^s + k^2 \bar{H}^s = j\omega\epsilon_0 \bar{M} - \frac{\nabla \nabla \cdot \bar{M}}{j\omega\mu_0} - \nabla \times \bar{J} \quad (2.10)$$

The equivalent sources $\bar{J}$ and $\bar{M}$ satisfy (2.1) and (2.2) where the $\bar{E} = \bar{E}^{tot}$ and $\bar{H} = \bar{H}^{tot}$.

To solve the Helmholtz equations (2.9) and (2.10) one could use different approaches. One way is to express the fields in terms of magnetic and electric vector potentials $\bar{A}$ and $\bar{F}$:

$$\bar{E}^s = \frac{\nabla \nabla \cdot \bar{A} + k^2 \bar{A}}{j\omega\epsilon_0} - \nabla \times \bar{F} \quad (2.11)$$

$$\overline{H}^s = \frac{\nabla \nabla \cdot \overline{F} + k^2 \overline{F}}{j\omega\mu_0} + \nabla \times \overline{A} \quad (2.12)$$

where the vector potentials satisfy:

$$\nabla^2 \overline{A} + k^2 \overline{A} = -\overline{J} \quad (2.13)$$

$$\nabla^2 \overline{F} + k^2 \overline{F} = -\overline{M} \quad (2.14)$$

A solution to these equations is:

$$\overline{A}(\overline{r}) = \iiint \overline{J}(\overline{r}') \frac{e^{-jk|\overline{r}-\overline{r}'|}}{4\pi|\overline{r}-\overline{r}'|} d\overline{r}' \quad (2.15)$$

$$\overline{F}(\overline{r}) = \iiint \overline{M}(\overline{r}') \frac{e^{-jk|\overline{r}-\overline{r}'|}}{4\pi|\overline{r}-\overline{r}'|} d\overline{r}' \quad (2.16)$$

Unfortunately, the above stated procedure requires an integration followed by a differentiation and is not well suited for numerical implementation. Usually the integrals cannot be evaluated in closed form and have to be evaluated at certain observation points using numerical quadrature. In this way the accuracy and the efficiency of computation will suffer.

One could also use a mixed potential formalism and search for a solution of the following form:

$$\overline{E}^s = -j\omega\mu_0 \overline{A} - \nabla \phi_e - \nabla \times \overline{F} \quad (2.17)$$

$$\overline{H}^s = -j\omega\epsilon_0 \overline{F} - \nabla \phi_m + \nabla \times \overline{A} \quad (2.18)$$

It can be shown that the magnetic vector potential, $\overline{A}$ and the electric vector potential, $\overline{F}$ satisfy the same equations (2.15) and (2.16) as before while the scalar magnetic potential ϕ_m and the scalar electric potential, ϕ_e are given by:

$$\phi_e = \frac{\rho_e}{\epsilon_0} \iiint \frac{e^{-jk|\overline{r}-\overline{r}'|}}{4\pi|\overline{r}-\overline{r}'|} d\overline{r}' \quad (2.19)$$

$$\phi_m = \frac{\rho_m}{\mu_0} \iiint \frac{e^{-jk|\overline{r}-\overline{r}'|}}{4\pi|\overline{r}-\overline{r}'|} d\overline{r}' \quad (2.20)$$

This approach provides a complete decoupling of the contributions to the radiated field from the electric current density and magnetic current density. Once the scalar and vector potentials are obtained the electromagnetic fields can be obtained using a single differentiation using (2.17) and (2.18). This approach is often preferred for numerical implementation.

2.2 Surface equivalence principle. Integral equations for perfectly conducting scatterers.

The surface equivalence principle considers the equivalent electric and magnetic sources to be on a closed surface coinciding with the scatterers surface. We will not present the most general form of it, but one that will be applied later to solve our problem. For further information on this, [14] is recommended.

Figure (2.2), (a) shows a particular form of the equivalence principle: Love's equivalent principle.

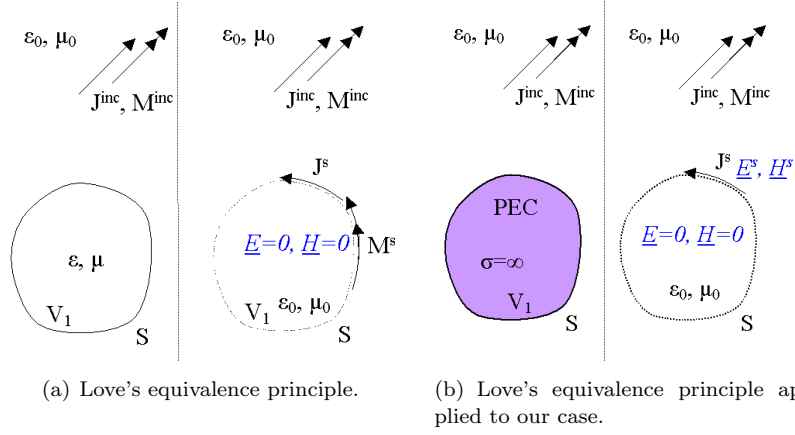


Figure 2.2:

The original problem is that of a scatterer of certain electric and magnetic properties, ϵ and μ which is irradiated by an incident electromagnetic field generated by some sources outside this scatterer, J^{inc} and M^{inc} . The scatterer is replaced by equivalent electric and magnetic sources which lie on a surface enclosing this scatterer. The boundary conditions which are needed to be satisfied by these sources are:

$$\overline{J}_s = \hat{n} \times [\overline{H}_{tot} - \overline{H}] \quad (2.21)$$

$$\overline{M}_s = -\hat{n} \times [\overline{E}_{tot} - \overline{E}] \quad (2.22)$$

These equivalent sources will radiate now in an unbounded space and will produce together with the incident fields the total field outside the scatterer. Since the fields inside the scatterer can be set to be anything the Love's Equivalence Principle assumes them to be zero so (2.21) and (2.22) become:

$$\overline{J}_s = \hat{n} \times \overline{H}_{tot} \quad (2.23)$$

$$\overline{M}_s = -\hat{n} \times \overline{E}_{tot} \quad (2.24)$$

In our case, figure (2.2), (b), the scatterer is a perfect electric conductor so the tangential component of the total electric field must vanish on the scatterers surface, meaning there can only exist electric surface currents. Applying Love's equivalence principle, only equivalent electric currents are needed since equation (2.24) becomes:

$$\overline{M}_s = 0 \quad (2.25)$$

In the final step we will combine the above boundary conditions set on the equivalent sources with the source-field relationships for this particular case of perfect electric conducting scatterer. Using (2.25) and (2.11) we get:

$$\hat{n} \times \overline{E}^{inc} = -\hat{n} \times \left(\frac{\nabla \nabla \cdot \overline{A} + k^2 \overline{A}}{j\omega\epsilon_0} \right), \quad r \in S, \quad \overline{A} = \overline{A}(\overline{J}_s) \quad (2.26)$$

which is one form of the Electrical Field Integral Equation (EFIE). This equation holds only for points on the surface S .

Using (2.23) and (2.12) we get:

$$\hat{n} \times \bar{H}^{inc} = \bar{J}_s - (\hat{n} \times \nabla \times \bar{A}), \quad r \in S^+, \quad \bar{A} = \bar{A}(\bar{J}_s) \quad (2.27)$$

the Magnetic Field Integral Equation(MFIE) which is enforced at an infinitesimal distance outside the scatterer surface(S^+).

Both equations could be used to solve for the unknown equivalent surface electrical currents and once they are determined, the electric and magnetic fields everywhere in space may be found using the source-field relations.

In our model we will use another idealization of the scatterer considering it to be a infinitesimally thin and perfect electric conductor plate. We could apply the surface equivalence principle in the same fashion as for the closed body scatterer. If the surface S is the scatterer surface, the current densities on both faces will collapse and the equation will not be able to distinguish between them. The solution will be the sum of these. Since the boundary condition for the electric field remains valid for infinitesimally thin structures we can still use the EFIE. But since the infinitesimally thin scatterer will not have the same magnetic field on both sides the boundary condition (2.23) is not valid any more so the MFIE, (2.27) cannot be used to compute the equivalent sources.

2.3 Numerical techniques: The moment method.

As stated above, the evaluation of integrals in the vector and scalar potentials is usually too complex to be given in closed form. Sometimes this is even impossible because the current distributions on the antenna are much too complicated. To overcome this impediment, approximations must be used. Numerical techniques can be used to solve these complicated problems. Since the chosen method here is the moment method a short background will be given, following [14].

The usual equation to be solved, as seen above, is of the form:

$$L \cdot f = g \quad (2.28)$$

where L is a continuous linear operator such as the integral operator or the differential operator. Function g represents the excitation, which is a known quantity while function f represents the unknown to be determined. If a unique solution exists to this problem it should be of the form:

$$f = L^{-1} \cdot g \quad (2.29)$$

where L^{-1} is the inverse operator. However, in practice L^{-1} cannot be determined so a numerical solution involving some sort of approximation is needed.

To understand better the above formulation we can rewrite equation 2.26 for the x -component as:

$$\left(\frac{\nabla \nabla + k^2}{j\omega\epsilon_0} \right) \cdot A_x(\bar{J}_s) = E_x^{inc} \quad (2.30)$$

Now it can be easily seen that:

$$L = \left(\frac{\nabla \nabla + k^2}{j\omega\epsilon_0} \right) A_x(\cdot)$$

$$f = \bar{J}_s$$

$$g = E_x^{inc}$$

The above linear operator L maps the unknown function f from its domain to the g function in its range. It is assumed the domain and range are different linear spaces. Without going to deep in linear algebra, we should have in mind the concept of inner product and basis functions set. Since in practice we are forced to project the functions of interest onto a finite-dimensional subspace of the original inner product space we can use a truncated basis function set to represent function f as:

$$f \approx f^N = \sum_{n=1}^N \alpha_n B_n \quad (2.31)$$

where α_n are scalar coefficients and B_n are basis functions of the finite subspace which don't have to be necessarily orthogonal!

The right hand side of equation 2.28 becomes:

$$L \cdot f = L \left(\sum_{n=1}^N \alpha_n B_n \right) = \sum_{n=1}^N \alpha_n (L \cdot B_n) \quad (2.32)$$

where LB_n can be expressed as:

$$LB_n \approx \sum_{m=1}^N l_{mn} T_m \quad (2.33)$$

The $T_1 \dots T_N$ functions form a basis function set in the g function space, called the testing functions set. That means we can also represent function g as:

$$g \approx g^N = \sum_{m=1}^N \beta_m T_m \quad (2.34)$$

Using (2.33) and (2.34) one can write the discretized form of (2.28) as:

$$\sum_{m=1}^N \sum_{n=1}^N l_{mn} \alpha_n T_m = \sum_{m=1}^N \beta_m T_m \quad (2.35)$$

which produces a system of $N \times N$ equations that will be solved for the scalar coefficients α_n :

$$\sum_{n=1}^N l_{mn} \alpha_n = \beta_m, \quad m = 1, 2, \dots, N \quad (2.36)$$

The β_m and l_{mn} scalar coefficients are computed as:

$$\beta_m = \langle T_m, g \rangle \quad (2.37)$$

$$l_{mn} = \langle T_m, LB_n \rangle \quad (2.38)$$

where $\langle, \rangle$ represents the inner product.

An important issue to be mentioned here is the choice of basis functions as they affect the desired accuracy of the approximate solution, the relative complexity of the resulting matrix entries, and computational constraints that place

an upper limit on the matrix size. Some examples of possible basis functions are: the dirac function, the pulse function, the sub-sectional triangle function or functions with higher order of differentiability as the quadratic spline function or the Lagrange polynomial functions. These are one-dimensional functions which can be extended to multi-dimensional basis functions. For the choice of testing function usually the dirac function is used resulting in the so called *Point Matching Technique*.

The differentiability constraints on the basis and testing functions should be also mentioned here. The specific linear operator to be discretized dictates the minimum polynomial degree for the basis and testing functions. In practice, this degree must increase in proportion to the number of derivatives operating on the unknown function. As stated in [14], page 201 to ensure a meaningful numerical solution or reasonable accuracy it is necessary to employ a combination of basis and testing functions that keep the above l_{mn} coefficients finite and well defined for any location of the basis or testing functions throughout the domain. This condition is also a minimum smoothness condition on representing the functions. For example if the l_{mn} coefficients involve a second order derivative one might employ:

- quadratic spline basis functions with Dirac delta testing functions
- triangle basis functions with pulse testing functions
- pulse basis functions with triangle testing functions
- Dirac delta basis functions with quadratic spline testing functions

Smother combinations would also satisfy the minimum criterium.

As a final remark, it has been shown in [14], page 205, table 5.2, that the impact of basis function choice is far more important than the choice of testing functions on the relative accuracy of the numerical solution.

A good article on the basis and testing functions employed using the moment method for different scattering problems is given by Glisson in [15].

Chapter 3

Implementation

3.1 Introduction.

The implementation described in this chapter follow Glisson [15] and Slättman [17].

First we should define the exact problem, the coordinate system and the specific equations used. As we are dealing with a antenna containing multiple conducting plates, we should first specify that the formulation will be restricted to flat perfect electric conducting plates/patches which are also considered to be infinitely thin. As figure (3.1) shows, the plates are lying in the xy plane and the multiple structure spans in the $+z$ direction.

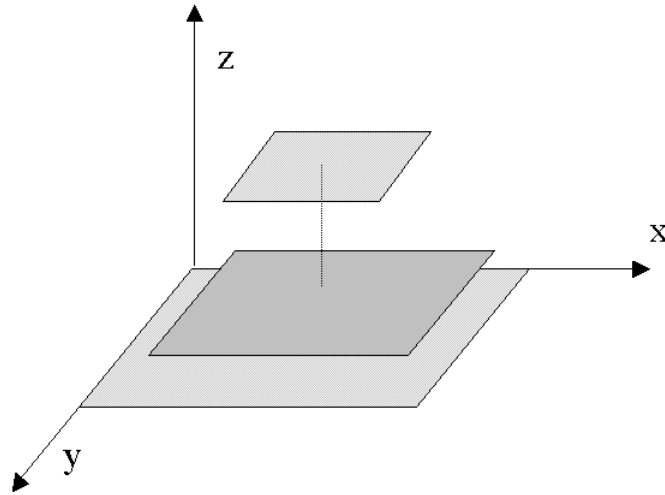


Figure 3.1: The multi-patch structure.

The patches may have any shape, but in anticipation of the rectangular grid we restricted them to rectangular or square patches. Off course, for any other shapes, the contour line of the patch will have to follow the grid providing a more or less accurate representation.

As we said before, the rectangular mesh has been adopted because of its simplicity of implementation. Also since, we restrict the geometry of one patch in 2 dimensions considering an infinitesimal thickness, there is no need for a more complex grid. The mesh is uniform and has square cells. Having a rectangular cell and a variable mesh thickness can be also implemented.

3.2 Currents model on the patches.

3.2.1 The EFIE formulation.

From the scattering problem, we know that the actual scatterer will be replaced with equivalent induced surface currents. As we consider them perfect electric conducting patches, only electric currents may exist.

The vector surface current density induced on the patch, may be described by its two components as follows:

$$\vec{J}(x, y) = J_x(x, y)\hat{x} + J_y(x, y)\hat{y} \quad (3.1)$$

The induced surface charge density ρ_s may be expressed through the continuity equation as:

$$\rho_s = -\frac{1}{j\omega} \vec{\nabla}_s \cdot \vec{J} = -\frac{1}{j\omega} \left[\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} \right] \quad (3.2)$$

In order to write the EFIE we refer again to equation (2.17), which becomes in our case:

$$\vec{E}^s = -j\omega\mu_0\vec{A} - \nabla\phi_e \quad (3.3)$$

The electric vector potential, $\vec{F}$ is zero since there is no magnetic current on the patch and the vector and scalar electric potentials are given by the same equations, (2.15) and (2.19). Applying the boundary condition on the patch surface the EFIE formulations can be written as a pair of coupled scalar equations:

$$j\omega\mu_0 A_x(\vec{r}) + \frac{\partial}{\partial x} \phi_e(\vec{r}) = E_x^{inc}(\vec{r}) \quad (3.4)$$

$$j\omega\mu_0 A_y(\vec{r}) + \frac{\partial}{\partial y} \phi_e(\vec{r}) = E_y^{inc}(\vec{r}) \quad (3.5)$$

These equations are valid only on the patches surface, $\vec{r} \in S$.

3.2.2 Basis functions

In choosing the basis function set we follow Glisson, [15] which introduces the so called *Rooftop Functions*. These functions as figure (3.2) shows are products between triangle function in the current flow direction and pulse function in the orthogonal direction.

These functions assume a linear variation in the current flow direction and in this sense they fulfill the continuity equation, (3.2), and model the charge distribution on the patch as piecewise constant within each cell. The *rooftop functions* are *Constant Normal-Linear Tangential* basis functions because they assure this kind of representation for the surface currents on the cell boundary, see [14], p. 395. At the structure edge the component of the normal current to

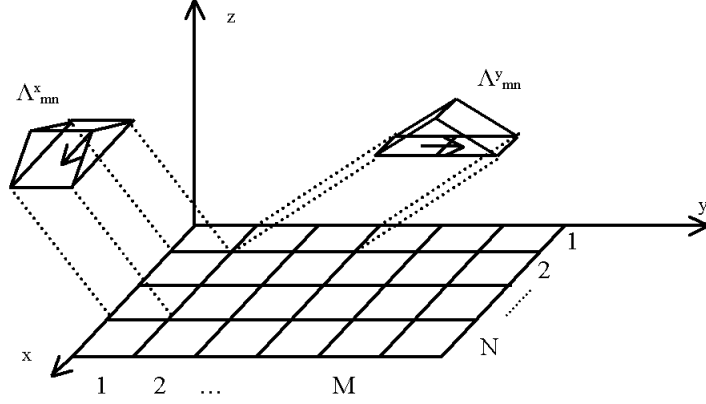


Figure 3.2: Current model. Basis functions.

the edge must be zero, so no basis function is needed there. The *rooftop basis functions* span across two cells and model the charge flow from one sub-domain to another.

We can introduce the x-directed and y-directed rooftop functions with the following notations:

$$\Lambda_{mn}^x(\bar{r}) = \Lambda_m(x) \cdot \Pi_n(y) \quad (3.6)$$

$$\Lambda_{mn}^y(\bar{r}) = \Pi_m(x) \cdot \Lambda_n(y) \quad (3.7)$$

where $\bar{r} = \bar{r}(x, y)$ and

$$\Lambda_i(p) = \begin{cases} 1 - \frac{p_i - p}{p_i - p_{i-1}}, & p_{i-1} < p < p_i \\ 1 - \frac{p - p_i}{p_{i+1} - p_i}, & p_i < p < p_{i+1} \\ 0, & \text{otherwise} \end{cases} \quad (3.8)$$

is the general one-dimensional triangle function and

$$\Pi_i(l) = \begin{cases} 1, & l_{i-1} < l < l_i \\ 0, & \text{otherwise} \end{cases} \quad (3.9)$$

is the one-dimensional pulse function.

The two components of the surface currents can be expressed than in terms of the rooftop functions and some scalar coefficients as:

$$J_x(\bar{r}) \approx \sum_{n=1}^N \sum_{m=1}^{M-1} J_{mn}^x \Lambda_{mn}^x(\bar{r}) \quad (3.10)$$

$$J_y(\bar{r}) \approx \sum_{n=1}^{N-1} \sum_{m=1}^M J_{mn}^y \Lambda_{mn}^y(\bar{r}) \quad (3.11)$$

where N is the number of mesh cells in x direction and M is the number of mesh cells in y direction.

3.3 Testing procedure

Following the differentiability conditions imposed on the testing functions and [15], the so called *Razor Blade* testing function is employed.

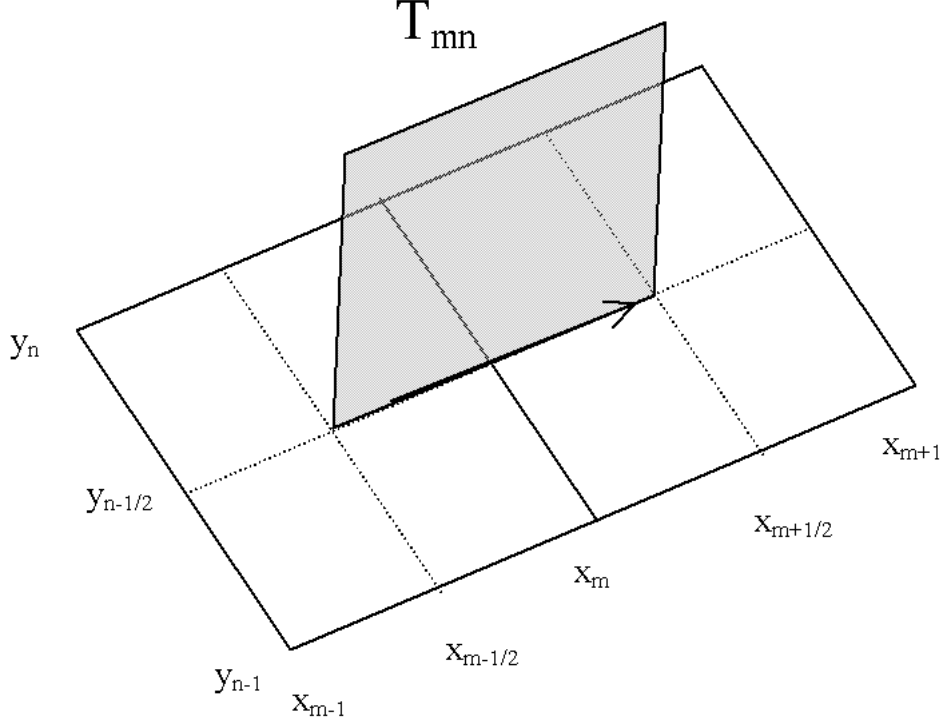


Figure 3.3: Test function for x-component of the electric field.

As seen in figure(3.3) the testing functions can be written as:

$$T_{mn}^x(\bar{r}) = \Pi_{m+1/2}(x)\delta(y - y_{n-1/2}) \quad (3.12)$$

for the x-component and

$$T_{mn}^y(\bar{r}) = \Pi_{n+1/2}(y)\delta(x - x_{m-1/2}) \quad (3.13)$$

for the y-component. The $\Pi_m(p)$ function is the one-dimension pulse function defined in 3.9 and $\delta(p - p_i)$ is the Dirac function.

The actual testing is done by integrating the two sides of equations (3.4) and (3.5) along the razor blade in the specified direction. This operation is nothing else than the inner product. This procedure yields the following equations:

$$j\omega\mu_0 \int_{x_{m-1/2}}^{x_{m+1/2}} A_x(x, y_{n-1/2})dx + \phi_e(x_{m+1/2}, y_{n-1/2}) - \phi_e(x_{m-1/2}, y_{n-1/2}) = \int_{x_{m-1/2}}^{x_{m+1/2}} E_x^{inc}(x, y_{n-1/2})dx \quad (3.14)$$

for $m = 1, 2, \dots, M-1$ and $n = 1, 2, \dots, N$ and

$$j\omega\mu_0 \int_{y_{n-1/2}}^{y_{n+1/2}} A_y(x_{m-1/2}, y) dy + \phi_e(x_{m-1/2}, y_{n+1/2}) - \phi_e(x_{m-1/2}, y_{n-1/2}) = \int_{y_{n-1/2}}^{y_{n+1/2}} E_y^{inc}(x_{m-1/2}, y) dy \quad (3.15)$$

for $m = 1, 2, \dots M$ and $n = 1, 2, \dots N - 1$. We have used the following integral to compute the testing on the scalar potential:

$$\int_{s_{i-1/2}}^{s_{i+1/2}} \frac{\partial \phi_e(s, p_j)}{\partial s} ds = \phi_e(s_{i+1/2}, p_j) - \phi_e(s_{i-1/2}, p_j) \quad (3.16)$$

To evaluate an approximate form of the integrals performed on the scalar potentials and the incident fields, we assume that they vary relatively slow on the integration domain, which can be achieved using fine meshing. Under this assumption we can approximate these integrals as follows:

$$\int_{s_{i-1/2}}^{s_{i+1/2}} F(s, p_j) ds = [F(s_{i-1/2}, p_j) + F(s_{i+1/2}, p_j)] \cdot \frac{[s_{i+1/2} - s_{i-1/2}]}{2} \quad (3.17)$$

Butler suggested in [16] to evaluate the function F only in one point, usually the center of the integration domain, but this was found to be too inaccurate in evaluating the vector potentials. As equation 3.17 shows, we chose to evaluate the function in minimum 2 points. The evaluation of the testing integral can be improved using quadrature formulas and more points.

With the above approximation (3.14) and (3.15) become:

$$j\omega\mu_0 [A_x(x_{m-1/2}, y_{n-1/2}) + A_x(x_{m+1/2}, y_{n-1/2})] \frac{\Delta x}{2} + \phi_e(x_{m+1/2}, y_{n-1/2}) - \phi_e(x_{m-1/2}, y_{n-1/2}) = E_x^{inc}(x_m, y_{n-1/2}) \Delta x \quad (3.18)$$

for $m = 1, 2, \dots M - 1$ and $n = 1, 2, \dots N$ and

$$j\omega\mu_0 [A_y(x_{m-1/2}, y_{n-1/2}) + A_y(x_{m-1/2}, y_{n+1/2})] \frac{\Delta y}{2} + \phi_e(x_{m-1/2}, y_{n+1/2}) - \phi_e(x_{m-1/2}, y_{n-1/2}) = E_y^{inc}(x_{m-1/2}, y_n) \Delta y \quad (3.19)$$

for $m = 1, 2, \dots M$ and $n = 1, 2, \dots N - 1$.

3.4 Matrix representation.

The resulting matrix equation is of the following form:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \cdot \begin{bmatrix} J_x \\ J_y \end{bmatrix} = \begin{bmatrix} V_x \\ V_y \end{bmatrix}$$

or, in a condensed manner:

$$[Z][J] = [V] \quad (3.20)$$

Since the EFIE is written in the form of two decoupled equations, the **A**, **B**, **C**, **D** matrixes contain the testing coefficients of the scattered electric field x and y components produced by the x and y current basis functions. J_x and J_y are the unknown current coefficients to be computed and V_x , V_y contain the testing coefficients of the incident electric field x and y components.

3.4.1 The A matrix

The **A** matrix elements represent the testing of the x component of the scattered field produced by the x-directed current basis functions and are denoted with

$$Z_{mn,pq}^{xx} = \langle T_{mn}^x | E^s(\Lambda_{mn}^x(x_p, y_q)) \rangle$$

The mn index denotes the tested razor blade coordinates, while the pq indexes the source basis function. The first superscript x denotes the testing direction while the second one, also x , the source current component.

Before given the actual expression of $Z_{mn,pq}^{xx}$, more explicit relations should be given for the vector and scalar potentials. The x component of the vector potential may be written as:

$$A_x(x_m, y_n) \approx \sum_{p=1}^{M-1} \sum_{q=1}^N J_{pq}^x A_{pq}^x(x_m, y_n) \quad (3.21)$$

where

$$\begin{aligned} A_{pq}^x(x_m, y_n) &= \iint \Lambda_{pq}^x(x_m, y_n) G(|\bar{r} - \bar{r}'|) dx' dy' = \\ &= \frac{1}{\Delta x} \cdot \left[\int_{x_{p+1}}^{x_p} \int_{y_{q-1}}^{y_q} (x' - x_{p-1}) G(|\bar{r} - \bar{r}'|) dx' dy' + \right. \\ &\quad \left. \int_{x_p}^{x_{p+1}} \int_{y_{q-1}}^{y_q} (x_{p+1} - x') G(|\bar{r} - \bar{r}'|) dx' dy' \right] \end{aligned} \quad (3.22)$$

and $G(|\bar{r} - \bar{r}'|)$ is Green's function:

$$G(|\bar{r} - \bar{r}'|) = \frac{e^{-jk\sqrt{(x_m-x')^2+(y_n-y')^2+(z-z')^2}}}{4\pi\sqrt{(x_m-x')^2+(y_n-y')^2+(z-z')^2}} \quad (3.23)$$

The y component of the vector potential can be written in a similar fashion.

The scalar potential may be written as:

$$\begin{aligned} \phi_e(x_m, y_n) &= \frac{j}{\omega\epsilon_0\mu_0} \iint \left(\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} \right) G(|\bar{r} - \bar{r}'|) dx' dy' \approx \\ &= \frac{j}{\omega\epsilon_0\mu_0} \cdot \sum_{p=1}^{M-1} \sum_{q=1}^N J_{pq}^x \left[\frac{\phi_{pq}(x_m, y_n)}{\Delta x} - \frac{\phi_{p+1,q}(x_m, y_n)}{\Delta x} \right] + \\ &= \frac{j}{\omega\epsilon_0\mu_0} \cdot \sum_{p=1}^M \sum_{q=1}^{N-1} J_{pq}^y \left[\frac{\phi_{pq}(x_m, y_n)}{\Delta y} - \frac{\phi_{p,q+1}(x_m, y_n)}{\Delta y} \right] \end{aligned} \quad (3.24)$$

where:

$$\phi_{pq}(x_m, y_n) = \iint \Pi_{pq}(x_m, y_n) G(|\bar{r} - \bar{r}'|) dx' dy' = \int_{x_{p-1}}^{x_p} \int_{y_{q-1}}^{y_q} G(|\bar{r} - \bar{r}'|) dx' dy' \quad (3.25)$$

with the above mentioned Green's function.

Now we are able to define the **A** matrix element as:

$$\begin{aligned} Z_{mn,pq}^{xx} = & j\omega\mu_0 [A_{pq}^x(x_{m-1/2}, y_{n-1/2}) + A_{pq}^x(x_{m+1/2}, y_{n-1/2})] \cdot \frac{\Delta x}{2} - \\ & \frac{j}{\omega\epsilon_0\mu_0} \left[\frac{\phi_{p+1,q}(x_{m+1/2}, y_{n-1/2})}{\Delta x} - \frac{\phi_{p,q}(x_{m+1/2}, y_{n-1/2})}{\Delta x} \right] - \\ & \frac{j}{\omega\epsilon_0\mu_0} \left[\frac{\phi_{p,q}(x_{m-1/2}, y_{n-1/2})}{\Delta x} - \frac{\phi_{p+1,q}(x_{m-1/2}, y_{n-1/2})}{\Delta x} \right] \end{aligned} \quad (3.26)$$

for $m, p = 1, 2, \dots, M-1$ and $n, q = 1, 2, \dots, N$

3.4.2 The B matrix

The **B** matrix elements represent the testing of the x component of the scattered field produced by the y-directed current basis functions and are denoted with:

$$Z_{mn,pq}^{xy} = \langle T_{mn}^x | E^s(\Lambda_{mn}^y(x_p, y_q)) \rangle$$

and are calculated according to:

$$\begin{aligned} Z_{mn,pq}^{xy} = & -\frac{j}{\omega\epsilon_0\mu_0} \left[\frac{\phi_{p,q+1}(x_{m+1/2}, y_{n-1/2})}{\Delta y} - \frac{\phi_{p,q}(x_{m+1/2}, y_{n-1/2})}{\Delta y} \right] - \\ & \frac{j}{\omega\epsilon_0\mu_0} \left[\frac{\phi_{p,q}(x_{m-1/2}, y_{n-1/2})}{\Delta y} - \frac{\phi_{p,q+1}(x_{m-1/2}, y_{n-1/2})}{\Delta y} \right] \end{aligned} \quad (3.27)$$

for $m = 1, 2, \dots, M-1$, $n = 1, 2, \dots, N$, $p = 1, 2, \dots, M$, $q = 1, 2, \dots, N-1$.

3.4.3 The C matrix

The **C** matrix elements represent the testing of the y component of the scattered field produced by the x-directed current basis functions and are denoted with:

$$Z_{mn,pq}^{yx} = \langle T_{mn}^y | E^s(\Lambda_{mn}^x(x_p, y_q)) \rangle$$

and are calculated according to:

$$\begin{aligned} Z_{mn,pq}^{yx} = & -\frac{j}{\omega\epsilon_0\mu_0} \left[\frac{\phi_{p+1,q}(x_{m-1/2}, y_{n+1/2})}{\Delta x} - \frac{\phi_{p,q}(x_{m-1/2}, y_{n+1/2})}{\Delta x} \right] - \\ & \frac{j}{\omega\epsilon_0\mu_0} \left[\frac{\phi_{p,q}(x_{m-1/2}, y_{n-1/2})}{\Delta x} - \frac{\phi_{p+1,q}(x_{m-1/2}, y_{n-1/2})}{\Delta x} \right] \end{aligned} \quad (3.28)$$

for $m = 1, 2, \dots, M$, $n = 1, 2, \dots, N-1$, $p = 1, 2, \dots, M-1$, $q = 1, 2, \dots, N$.

3.4.4 The D matrix

The **D** matrix elements represent the testing of the y component of the scattered field produced by the y-directed current basis functions and are denoted with

$$Z_{mn,pq}^{yy} = \langle T_{mn}^y | E^s(\Lambda_{mn}^y(x_p, y_q)) \rangle$$

which is written as:

$$\begin{aligned} Z_{mn,pq}^{yy} = & j\omega\mu_0 [A_{pq}^y(x_{m-1/2}, y_{n-1/2}) + A_{pq}^y(x_{m-1/2}, y_{n+1/2})] \cdot \frac{\Delta y}{2} - \\ & \frac{j}{\omega\epsilon_0\mu_0} \left[\frac{\phi_{p,q+1}(x_{m-1/2}, y_{n+1/2})}{\Delta y} - \frac{\phi_{p,q}(x_{m-1/2}, y_{n+1/2})}{\Delta y} \right] - \\ & \frac{j}{\omega\epsilon_0\mu_0} \left[\frac{\phi_{p,q}(x_{m-1/2}, y_{n-1/2})}{\Delta y} - \frac{\phi_{p,q+1}(x_{m-1/2}, y_{n-1/2})}{\Delta y} \right] \end{aligned} \quad (3.29)$$

for $m, p = 1, 2, \dots, M$ and $n, q = 1, 2, \dots, N - 1$

3.5 Numerical integration. Efficiency considerations.

As saw above, to evaluate the elements of the **[Z]** matrix in (3.20) surface integrals have to be performed. These have to be evaluated using numerical quadrature, in our case Gauss-Legendre quadrature is used.

At a closer look to the **[Z]** elements to be evaluated, one can see a case where the testing and the source overlaps. This is a special case since the function to be integrated contains a singularity. To overcome this problem one could extract the singularity out of the integral as suggested in [14] or perform a coordinate change to polar coordinates where the singularity vanishes [17]. The latter has been implemented here.

Without going to much in details, the evaluation of the self terms is performed according to:

$$\begin{aligned} \int_{-u_1}^{u_1} \int_{-v_1}^{v_1} \frac{e^{-jk\sqrt{u^2+v^2}}}{4\pi\sqrt{u^2+v^2}} dudv = & \frac{j}{\pi k} \left[\int_0^{\alpha_0} [e^{-j\frac{ku_1}{\cos\alpha}} - 1] d\alpha + \right. \\ & \left. \int_{\alpha_0}^{\pi/2} [e^{-j\frac{kv_1}{\sin\alpha}} - 1] d\alpha \right], \quad \alpha_0 = \arctan\left(\frac{v_1}{u_1}\right) \end{aligned} \quad (3.30)$$

for the scalar potential, and:

$$\begin{aligned} \int_{-u_1}^{u_1} \int_{-v_1}^{v_1} (u + \Delta u) \frac{e^{-jk\sqrt{u^2+v^2}}}{4\pi\sqrt{u^2+v^2}} dudv = & \frac{1}{2\pi k \Delta u} \cdot \\ \int_0^\pi [(jku_1 + jkR(\alpha)\cos\alpha + \cos\alpha)e^{-jkR(\alpha)} - jku_1 - \cos\alpha] d\alpha \end{aligned} \quad (3.31)$$

where:

$$R(\alpha) = \begin{cases} \frac{u_1}{\cos\alpha}, & 0 \leq \alpha < \arctan\left(\frac{v_1}{u_1}\right) \\ \frac{v_1}{\sin\alpha}, & \arctan\left(\frac{v_1}{u_1}\right) \leq \alpha < \pi - \arctan\left(\frac{v_1}{u_1}\right) \\ -\frac{u_1}{\cos\alpha}, & \pi - \arctan\left(\frac{v_1}{u_1}\right) \leq \alpha < \pi \end{cases}$$

and:

$$\int_{-u_1}^{u_1} \int_{-v_1}^{v_1} (\Delta u - u) \frac{e^{-jk\sqrt{u^2+v^2}}}{4\pi\sqrt{u^2+v^2}} dudv = \frac{1}{2\pi k \Delta u} \cdot \int_0^\pi [(jku_1 - jkR(\alpha)\cos\alpha - \cos\alpha)e^{-jkR(\alpha)} - jku_1 - \cos\alpha] d\alpha \quad (3.32)$$

where:

$$R(\alpha) = \begin{cases} \frac{u_1}{\cos\alpha}, & 0 \leq \alpha < \arctan(\frac{v_1}{u_1}) \\ \frac{v_1}{\sin\alpha}, & \arctan(\frac{v_1}{u_1}) \leq \alpha < \pi - \arctan(\frac{v_1}{u_1}) \\ -\frac{u_1}{\cos\alpha}, & \pi - \arctan(\frac{v_1}{u_1}) \leq \alpha < \pi \end{cases}$$

for the vector potentials.

The evaluation of double integrals is very time consuming so to improve the computational efficiency the fact that the **[A]**, **[B]**, **[C]**, **[D]** matrixes are *Block Toeplitz* has been used. This property is a consequence of using equal-size rectangular cells and because of the translation symmetry in the plate geometry. Thus, to compute the whole matrixes only the first line or first column is needed providing valuable time saving in filling the matrixes. To minimize the calculation time further, the quadrature process has been adapted to work with matrix input. In this way, all elements are computed at once and not one by one. As the self terms need special treatment, they are given some value in the first step of calculation which involves matrix-wise operations and than the correct values are inserted.

The whole implementation is done using **Matlab** environment. The solution to the matrix equation (3.20) can be written as:

$$[J] = [Z]^{-1} \cdot [V] \quad (3.33)$$

The inversion of the **[Z]** matrix is handled by **Matlab**'s internal algorithms, which are very efficient.

3.6 Excitation options.

Excitation options refer to the vector **[V]** evaluation. It's elements contain the testing of the incident fields on the patch surface.

3.6.1 Plane wave excitation.

The plane wave excitation considers a normal incident, x and/or y polarized plane wave impinging on the patches from the positive z direction. This is modelled by having a constant incident field excitation in all the cells, so the elements of the **[V]** are computed according to:

$$V_x(x_m, y_n) = \int_{L_x} E_x^{inc}(u, y_{n-1/2}) du = E_x^{inc}(x_m, y_{n-1/2}) \cdot \Delta x \quad (3.34)$$

and:

$$V_y(x_m, y_n) = \int_{L_y} E_y^{inc}(x_{m-1/2}, u) du = E_y^{inc}(x_{m-1/2}, y_n) \cdot \Delta y \quad (3.35)$$

for $m = 1, 2, \dots, M-1$ and $n = 1, 2, \dots, N$. L_x and L_y are the integration lines defined by the testing functions, i.e. the razor blade functions.

3.6.2 Probe excitation.

A typical probe excitation is depicted in figure 3.4. The probe is modelled as

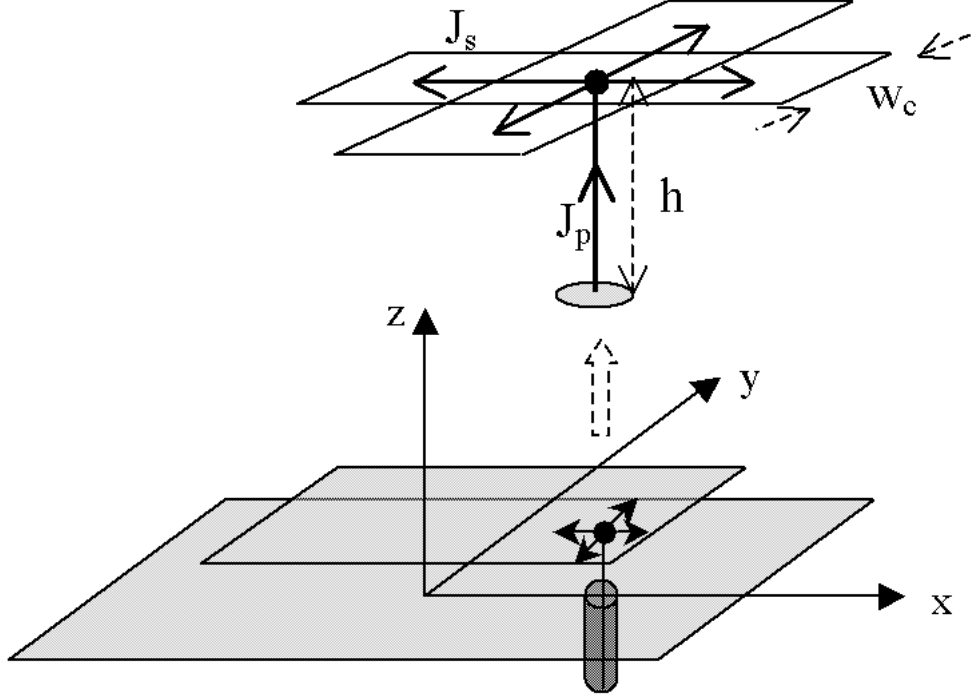


Figure 3.4: Two patches structure with probe feed.

a small constant current that feeds the excited patch currents. The excitation vector $[\mathbf{V}]$ has two different components. The first component is the near field of the probe which will irradiate the patches cells. The second component is the currents which flow on the patch from the probe. We will discuss both.

The probe feeds the patch with currents that will flow out from the feeding point in all directions. In our case, since we compute the $\mathbf{x}$ and $\mathbf{y}$ components of the surface currents, we choose to set four currents as depicted in the previous picture to have equal amplitudes and signs so that they flow out from the feeding point. In order to be consistent with Kirchoff's current continuity law, the probe current is computed as:

$$I_p = 4 \cdot J_s \cdot w_c \quad (3.36)$$

where $J_s = 1[A/m]$ is the amplitude of the closest surface current (i.e. closest rooftop function) flowing on the patch from the probe and w_c is the cell width.

We have to come back now to the probe model. The probe is modelled as an infinitesimally thin constant current of length h with:

$$\bar{\mathbf{I}} = \hat{\mathbf{z}} I_p \quad (3.37)$$

placed between the two patches. The near field components radiated by it are given by [1]:

$$E_r^{inc} = \eta \frac{I_p h \cos \theta}{2\pi r^2} \left[1 + \frac{1}{jkr} \right] e^{-jkr} \quad (3.38)$$

$$E_{\theta}^{inc} = j\eta \frac{kI_p h \sin\theta}{4\pi r} \left[1 + \frac{1}{jkr} - \frac{1}{(kr)^2} \right] e^{-jkr} \quad (3.39)$$

Using the above, the excitation vector $[\mathbf{V}]$ is computed. It is clear that this model has simplified the current probe to a constant phase current but as it will be shown in chapter (4) the simulation results are in very good agreement with the measurements. The aperture feed is another excitation type used frequently in practical antennas.

3.7 The multi-patch problem.

So far, we haven't mentioned anything about the $(z - z')^2$ term in the Green's function. It represents the spacing in z direction between the tested domain and the source basis function. For the case of one patch, this term was always zero. In the case of multi-patch structure this term is the actual distances between patches.

The $[\mathbf{Z}]$ matrix can be formed in many ways for the multi-patch structure. In our case we'll take the example of two patches to explain it, while the general case follows from it.

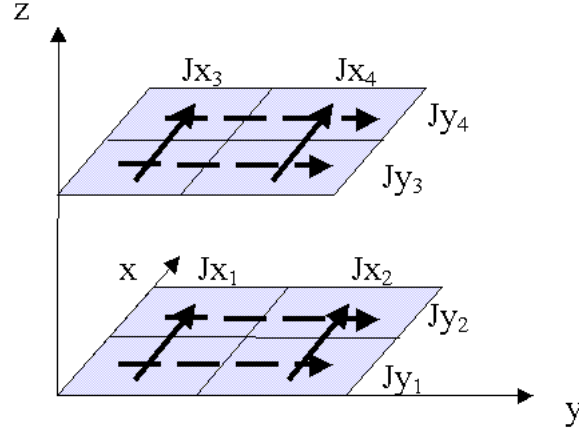


Figure 3.5: Two patches structure.

Figure (3.5) shows a two patch structure of 2×2 cells. The x and y directed induced currents are depicted using only one index. Using the testing notations introduced before, the detailed $[\mathbf{Z}]$ matrix is written as follows:

$$\begin{bmatrix} \langle T_{x1}|E^s(J_{x1}) \rangle & \dots & \langle T_{x1}|E^s(J_{x4}) \rangle & \langle T_{x1}|E^s(J_{y1}) \rangle & \dots & \langle T_{x1}|E^s(J_{y4}) \rangle \\ \vdots & \dots & \vdots & \vdots & \dots & \vdots \\ \langle T_{x4}|E^s(J_{x1}) \rangle & \dots & \langle T_{x4}|E^s(J_{x4}) \rangle & \langle T_{x4}|E^s(J_{y1}) \rangle & \dots & \langle T_{x4}|E^s(J_{y4}) \rangle \\ \langle T_{y1}|E^s(J_{x1}) \rangle & \dots & \langle T_{y1}|E^s(J_{x4}) \rangle & \langle T_{y1}|E^s(J_{y1}) \rangle & \dots & \langle T_{y1}|E^s(J_{y4}) \rangle \\ \vdots & \dots & \vdots & \vdots & \dots & \vdots \\ \langle T_{y4}|E^s(J_{x1}) \rangle & \dots & \langle T_{y4}|E^s(J_{x4}) \rangle & \langle T_{y4}|E^s(J_{y1}) \rangle & \dots & \langle T_{y4}|E^s(J_{y4}) \rangle \end{bmatrix} =$$

$$\begin{bmatrix} A_{11} & A_{12} & B_{11} & B_{12} \\ A_{21} & A_{22} & B_{21} & B_{22} \\ C_{11} & C_{12} & D_{11} & D_{12} \\ C_{21} & C_{22} & D_{21} & D_{22} \end{bmatrix}$$

The second form is a more succinct block matrix formulation. The sub-matrixes indexation has the following meaning: the first index represents the tested patch and the second one represents the source patch. In this way contributions between every patch can be separated and calculated using the same procedures in the section 3.4. One could see that evaluation of the reaction integrals between two different patches is somewhat simplified since there is no self term involved.

Following the above, the general multi-patch matrix equation may be written:

$$\begin{bmatrix} A_{11} & \dots & A_{1N} & B_{11} & \dots & B_{1N} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{N1} & \dots & A_{NN} & B_{N1} & \dots & B_{NN} \\ C_{11} & \dots & C_{1N} & D_{11} & \dots & D_{1N} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ C_{N1} & \dots & C_{NN} & D_{N1} & \dots & D_{NN} \end{bmatrix} \cdot \begin{bmatrix} J_{x1} \\ \vdots \\ J_{xN} \\ J_{y1} \\ \vdots \\ J_{yN} \end{bmatrix} = \begin{bmatrix} V_{x1} \\ \vdots \\ V_{xN} \\ V_{y1} \\ \vdots \\ V_{yN} \end{bmatrix}$$

where N represents the number of patches. The matrix equation is solved in the same manner as explained above.

3.8 Infinite PEC ground plane problem.

Because infinite perfect electric conductor ground plane is a useful tool, it has implemented to check the influence of reflector patches. It also provides a way of checking convergence for different reflector patch sizes. Consider figure(3.6), where two patches are considered lying above an infinite PEC GP.

Using *Image Theory*, see [1], p. 165, we can construct an equivalent problem. As we know the image electric images of the currents have the same amplitude but are with opposite sign. The Jx_i and Jy_i represent all the x-directed respectively y-directed currents on patch i , figure 3.6. In case of a plane wave incidence excitation, $\underline{E}^{inc}$, coming from the +z direction, there should be also an image plane wave, $\underline{E}^{im}$ coming from -z direction. One could understand this better if one considers the plane wave to have a source current very far from the patch. According to *Image Theory* that source should also have an image which is radiating the image plane wave.

Our initial bounded problem has been in this way transposed into a multi-patch problem in free space. This can be solved as seen in the previous chapter, with the note that the number of unknowns is still equal to the initial problem, although now the number of patches has doubled.

After some algebraic manipulations, the matrix equation for the above case

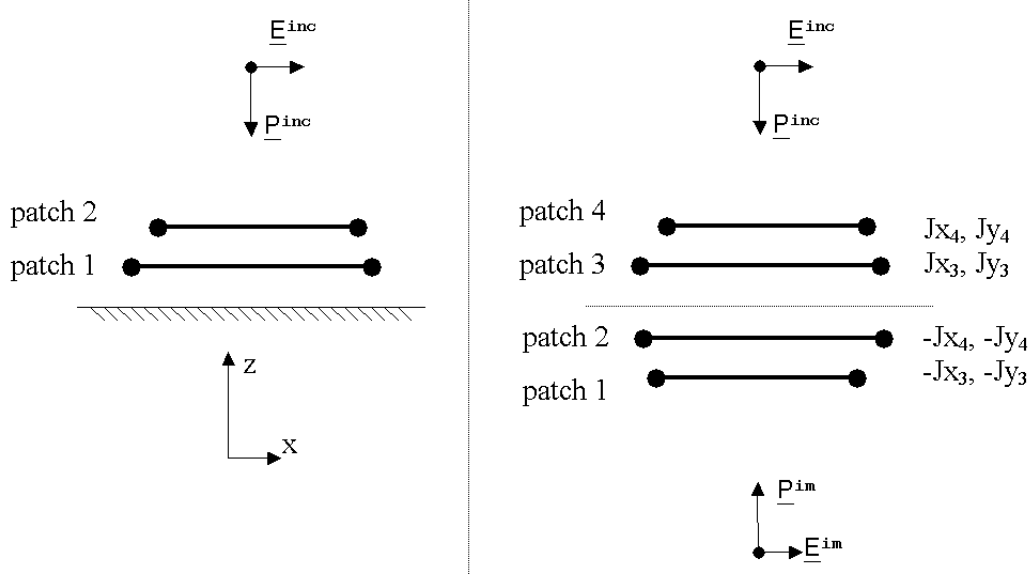


Figure 3.6: Two patches above infinite ground plane and equivalent model using Image Theory.

can be written as:

$$\begin{bmatrix} A_{23} - A_{22} & A_{24} - A_{21} & B_{23} - B_{22} & B_{24} - B_{21} \\ A_{13} - A_{12} & A_{14} - A_{11} & B_{13} - B_{12} & B_{14} - B_{11} \\ C_{23} - C_{22} & C_{24} - C_{21} & D_{23} - D_{22} & D_{24} - D_{21} \\ C_{13} - C_{12} & C_{14} - C_{11} & D_{13} - D_{12} & D_{14} - D_{11} \end{bmatrix} \cdot \begin{bmatrix} J_{x3} \\ J_{x4} \\ J_{y3} \\ J_{y4} \end{bmatrix} = \begin{bmatrix} V_{x3} \\ V_{x4} \\ V_{y3} \\ V_{y4} \end{bmatrix}$$

In conclusion the infinite ground plane problem will affect only the filling time of the $[\mathbf{Z}]$ matrix, which doubles, since the number of patches has doubled, but the final matrix equation has the same complexity in terms of number of unknowns as the initial problem without the ground plane. The above procedure can be easily generalized for N patches above ground plane.

3.9 Far field calculation.

After computing the induced surface currents on the patches, we are now able to calculate the total field according to equation (2.5). As the incident field is a given quantity we need to compute only the scattered field produced by the equivalent surface currents on the patch. In order to be able to apply the far field amplitude and phase approximations and so to reduce complexity in evaluating the scattered field, the cell should be smaller than $\frac{\lambda}{10}$.

Using the source-field relations we can compute the magnetic vector potentials. Since the current sources have only x and y components the magnetic vector potentials can be written as:

$$\bar{A}(r, \theta, \phi) = \mu_0 \iint \bar{J}(x', y', z') G(|\bar{r} - \bar{r}'|) dx' dy' =$$

$$\frac{\mu_0}{4\pi} \cdot \frac{e^{-jk r}}{r} \iint \bar{J}(x', y', z') e^{jk(\sin \theta \cos \phi x' + \sin \theta \sin \phi y' + \cos \theta z')} dx' dy' \quad (3.40)$$

Considering the far field evaluation at distance $r=1[m]$ from the source, we have:

$$\begin{aligned} \bar{A}(r, \theta, \phi) &\approx \frac{\mu_0}{4\pi} \sum_{m=1}^M \sum_{n=1}^N \bar{J}_{mn}(x', y', z') e^{jk(\sin \theta \cos \phi x' + \sin \theta \sin \phi y' + \cos \theta z')} \Delta x' \Delta y' = \\ &\frac{\mu_0}{4\pi} \sum_{m=1}^M \sum_{n=1}^N \bar{J}_{mn}^x(x', y', z') e^{jk(\sin \theta \cos \phi x' + \sin \theta \sin \phi y' + \cos \theta z')} \Delta x' \Delta y' + \\ &\frac{\mu_0}{4\pi} \sum_{m=1}^M \sum_{n=1}^N \bar{J}_{mn}^y(x', y', z') e^{jk(\sin \theta \cos \phi x' + \sin \theta \sin \phi y' + \cos \theta z')} \Delta x' \Delta y' = \\ &\bar{A}_x(r, \theta, \phi) + \bar{A}_y(r, \theta, \phi), \quad r = 1 \end{aligned} \quad (3.41)$$

The vector potential calculated above are than transferred to spherical coordinates and using the far field approximations the electric far field radiation is evaluated as:

$$\bar{E}_r(r, \theta, \phi) \approx 0 \quad (3.42)$$

$$\bar{E}_\theta(r, \theta, \phi) \approx -j\omega \bar{A}_\theta(r, \theta, \phi) \quad (3.43)$$

$$\bar{E}_\phi(r, \theta, \phi) \approx -j\omega \bar{A}_\phi(r, \theta, \phi) \quad (3.44)$$

Chapter 4

Verifications

4.1 Verification with literature.

In this chapter we will give the simulation results on different structures. The first three sections will provide known results for standard problems which may be compared with results from literature. The next two sections will study the convergence problem, in the far field and for the currents on the patches for different structures.

4.1.1 1λ patch in free space.

This is a classical problem referred also by Peterson in [14], page 424. It consists of a square patch of $\lambda \times \lambda$ size placed in an unbounded homogenous environment, free space. The excitation is considered to be a normal incident plane wave with $E_x = 1[\frac{V}{m}]$ amplitude, coming from $+z$ direction. As we said in the previous chapters, the patches will always lay in the xy plane.

The mesh used for this case had 10×10 cells providing a total of 90 basis functions for each surface current component. Numerical values are given in appendix ?? for a smaller structure. Comparison data was provided by Andrew F. Peterson. The matrix filling time was approximately 7 sec. while inverting time is negligible.

Figures 4.1 and 4.2 show the 3D plots of the surface currents on the patch.

For a better comparison with [14] we present two cuts along the center of the patch for the J_x current component.

One should see that the dominant component of the surface current is as expected the J_x which is ≈ 10 times larger than the cross-polarized component. It also presents the expected standing wave pattern, while the cuts can be compared with current distribution induced on thin strips excited by incident TE and TM plane waves. Regarding the cross-polar equivalent current, it's only excitation is the dominant current, J_x and it has a antisymmetrical pattern.

4.1.2 Dipole current distribution.

The current distribution on a dipole has been subject of many studies and it provides a reference when verifying moment method programs. In this case,

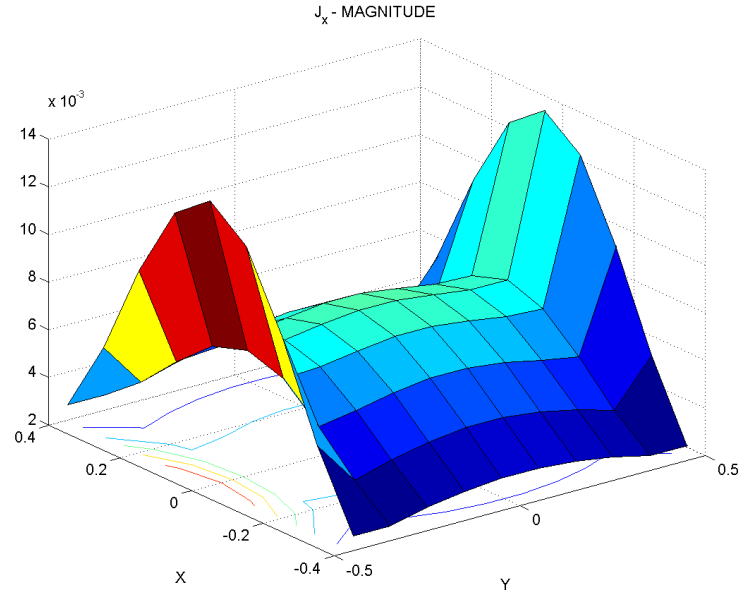


Figure 4.1: J_x surface current component magnitude for a $\lambda \times \lambda$ patch in free space. x-polarized, normal incidence, plane wave.

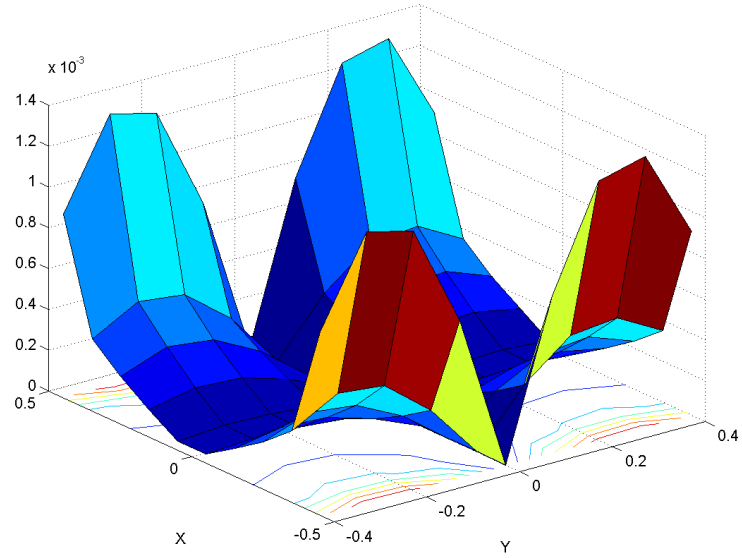


Figure 4.2: J_y surface current component magnitude for a $\lambda \times \lambda$ patch in free space. x-polarized, normal incidence, plane wave.

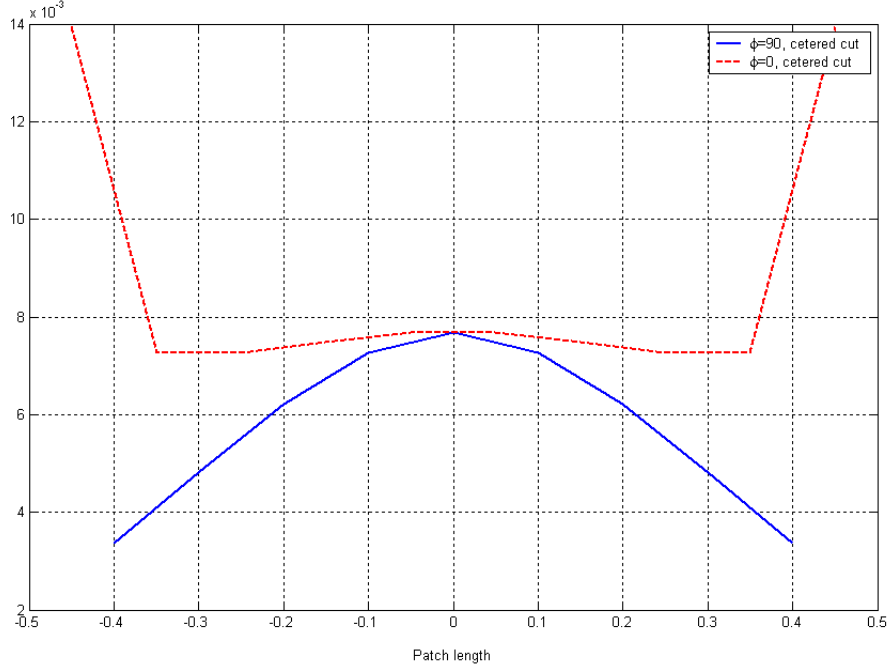


Figure 4.3: J_x surface current component cuts along the center of the patch for $\phi = 0$ and $\phi = 90$.

the simulator has been modified to simulate plane strips of given width. The strips are lying in the xy plane and the linear array spans in the $+z$ direction. The mesh contains only one cell in the y direction. In this way we can simulate different dipoles and also the Yagi-Uda array. The equivalent surface currents can have only a x component, which is true for the real case if we keep the strips sufficiently thin.

Figure 4.4 and 4.5 show the equivalent current distribution on $\frac{\lambda}{2}$ and λ long strips when the center current basis function is given amplitude of 1[A]. The widths were chosen so that results can be compared to [1, Fig.8.13], where the author uses cylindrical dipoles of radius: 10^{-3} and 10^{-4} .

4.1.3 Yagi-Uda arrays.

As said before the program can be used to analyze linear arrays of dipoles and so the Yagi-Uda array. The following example considers a six elements Yagi-Uda antenna with parameters given in table 4.1.

The normalized amplitude antenna pattern is given in figure 4.6. An extensive optimization analysis has been done in [2] for this array and results are also presented in [1], page 526. Figure 4.6 shows the far field patterns generated by the MoM simulator and the one presented by Chen in [2]. For further comparison between Chen's results and MoM simulator, we present in table 4.2, the normalized current amplitudes and phase in the elements of the antenna.

As it can be seen, the current amplitudes are a little bit different although

-	length[λ]	spacing[λ]
reflector	0.51	-
excited	0.49	0.25
director 1	0.43	0.31
director 2	0.43	0.31
director 3	0.43	0.31
director 4	0.43	0.31

Table 4.1: Yagi-Uda array parameters.

element	MoM simulator		Chen	
	Amp(Phase)	Ph. shift	Amp(Phase)	Ph. shift
reflector	0.59(76°)	-	-	-
excited	1(-39°)	115°	1(-39°)	-
director 1	0.6(-209.4°)	170.4°	0.67(-205°)	166
director 2	0.55(-317°)	107.6°	0.93(-324°)	119°
director 3	0.7(-451°)	134.6°	1.121(-475°)	151°
director 4	0.56(-595°)	143.4°	0.8(-612°)	137°

Table 4.2: Initial 6-element Yagi-Uda array.

the phases are quite close. This difference can be explained by the fact that Chen uses a three term expression with complex coefficients to approximate

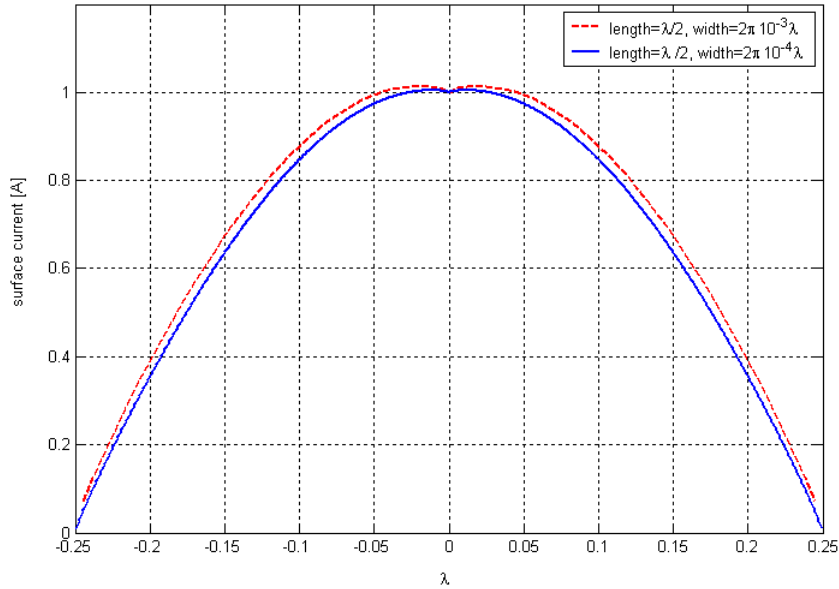


Figure 4.4: Current distribution on a thin $\frac{\lambda}{2}$ strip, middle current excitation of 1[A].

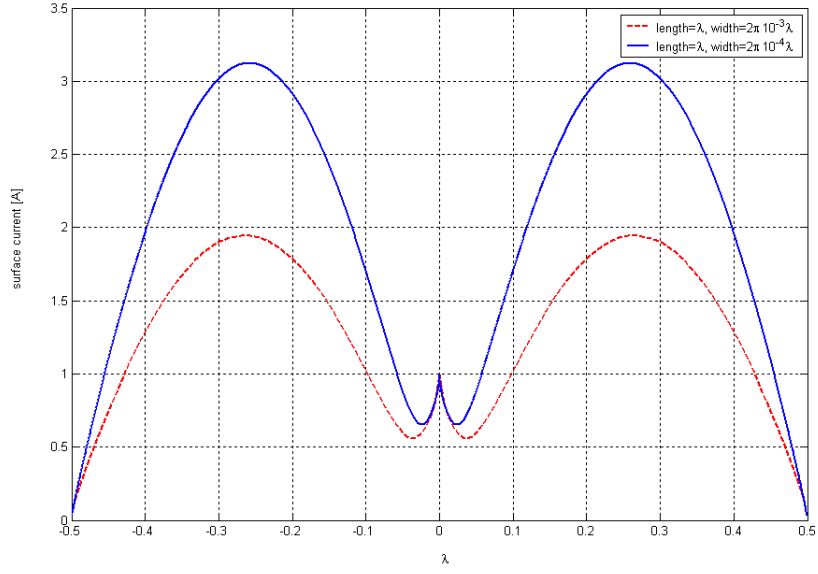


Figure 4.5: Current distribution on a thin λ strip, middle current excitation of 1[A].

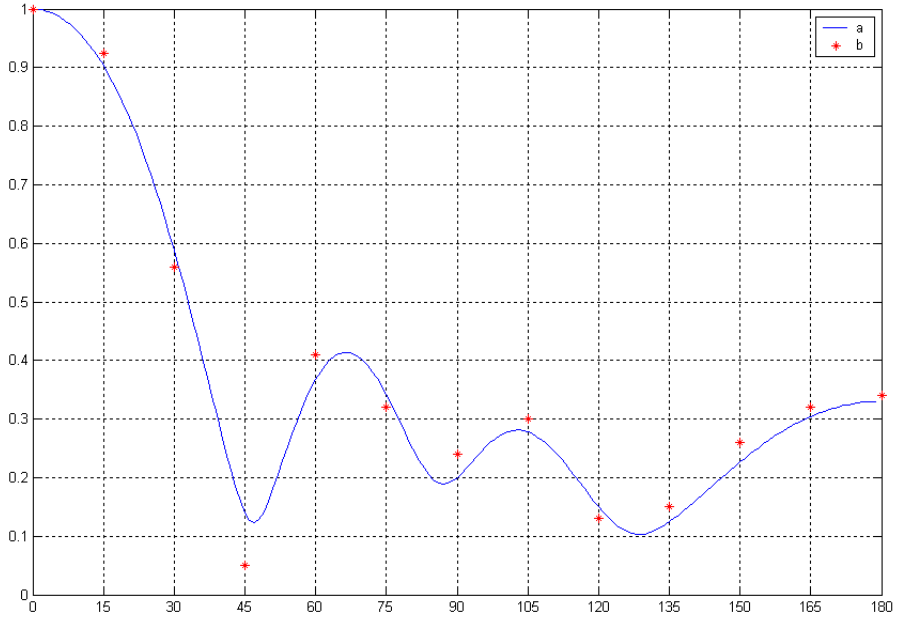


Figure 4.6: $E_{abs}(\phi = 90^\circ)$, normalized amplitude pattern of 6-elements, uniformly spaced Yagi-Uda array. (a) MoM (b) Chen

the current distributions in the dipoles while our method is more close to reality. The excitation of the driven element, is done considering the center basis

function to be excited by a equivalent voltage source of 1[V]. This is in some-way similar to the delta gap source modelling. Also we have to remember that Chen's model uses cylindrical dipoles of diameter $a = 0.003369\lambda$ while in our case we consider strips of 0.01λ in width and infinitely thin. The difference in the far field pattern is the result of the difference between the currents for the reasons showed above.

In his article, [2], Chen performs an optimization on this array using a perturbation method of director spacings and all element lengths. It is also important to look at the results of this analysis and more important to the conclusions. For this we included a table of currents for the optimum array obtained from the MoM simulator and from Chen's article:

element	MoM simulator		Chen	
	Amp(Phase)	Ph. shift	Amp(Phase)	Ph. shift
reflector	0.41(125.8°)	-	-	-
excited	1(-31°)	156.8°	1(-31°)	-
director 1	0.73(-188.14°)	157.14°	0.91(-207°)	176°
director 2	0.64(-345.8°)	157.66°	0.73(-380°)	173°
director 3	0.58(-508.71°)	162.9°	0.58(-547°)	167°
director 4	0.35(-669.43°)	160.72°	0.30(-708°)	161°

Table 4.3: Optimized 6-element Yagi-Uda array.

Looking at the above tables, one can see that the currents amplitudes and phase fluctuate widely in the initial design. The results of the optimized Yagi-Uda, from [2], page 14, and included above show that the currents amplitudes decrease smoothly and rapidly and the phases shift between elements tends to remain constant as one should expect. This optimization seams to lead to an array on which a travelling wave with a constant phase velocity is sustained. One could also argue of the accuracy of the results since Thiele in [8] shows that optimum directivity for Yagi-Uda array implies that the phase velocity of the surface wave should be smoothly decreasing. Nevertheless, this conclusion is very important and we will try to see if it can be applied to the multi-patch structure optimization. Figure 4.7 shows the normalized amplitude pattern for the optimized array.

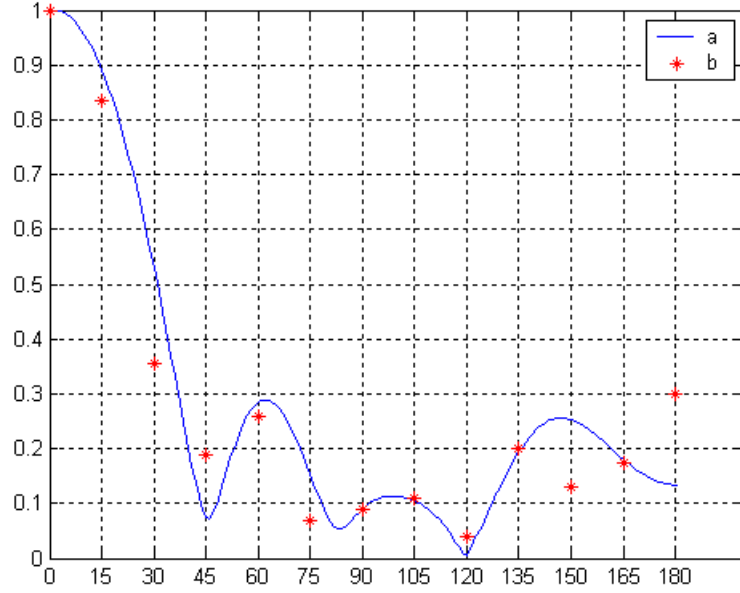


Figure 4.7: $E_{abs}(\phi = 90^\circ)$, normalized amplitude pattern of 6-elements, optimized Yagi-Uda array. (a) MoM (b) Chen

4.1.4 Solution convergence.

In order to observe the solution convergence the following two patch structure depicted in figure 4.8 is analyzed. We will show that even for a coarse mesh we can achieve a good solution for the radiated far field.

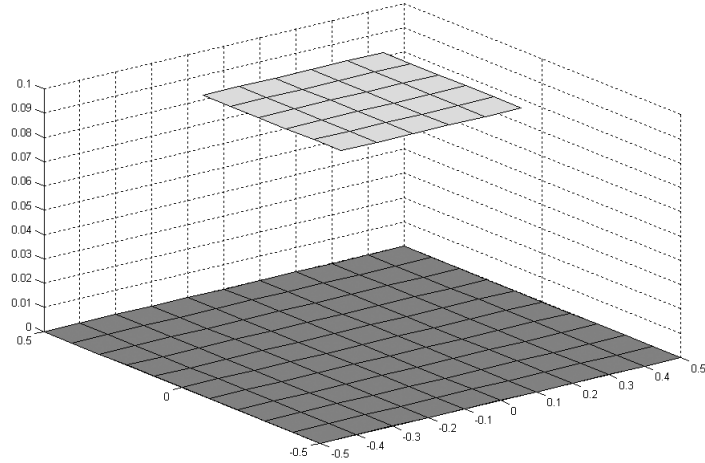


Figure 4.8: Two patch structure in free space.

The first patch is $\lambda \times \lambda$ in size, while the second one is $\frac{\lambda}{2} \times \frac{\lambda}{2}$ in size. They are spaced at $\frac{\lambda}{10}$. The second patch is excited by setting a centered current x-

directed basis function coefficient to 1[A]. An initial analysis with a grid cell of $\frac{\lambda}{10}$ for both patches is done, followed by mesh refinement to $\frac{\lambda}{20}$. The convergence for the far field is checked for $\phi = 0^\circ$ and $\phi = 90^\circ$ cuts in figure 4.9.

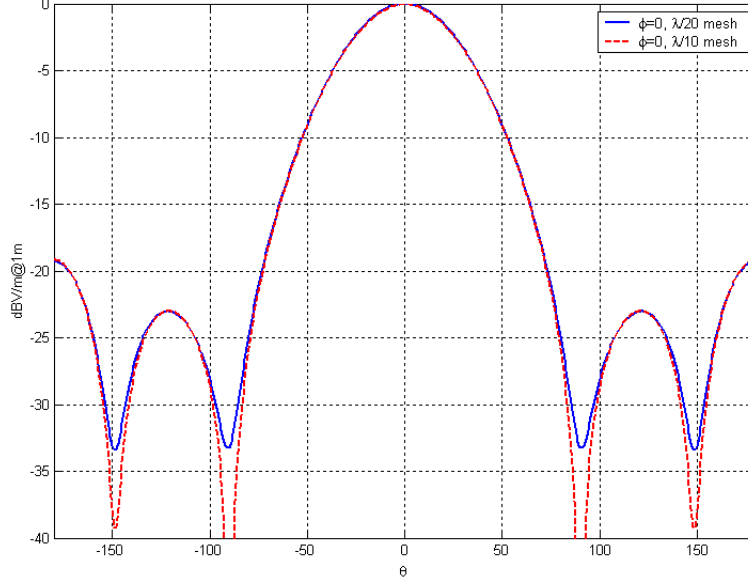


Figure 4.9: Convergence of the scattered far field for, $E_\theta(\phi = 0)$. (a) $\frac{\lambda}{10}$ mesh (b) $\frac{\lambda}{20}$ mesh.

As it can be seen even for a mesh of $\frac{\lambda}{10}$, which considers only 5×5 cells for the small patch, the results are accurate. The far field convergence is very good.

4.1.5 Solution consistence.

As we mentioned before, the infinite perfect electric conducting ground plane was modelled here with the use of image theory. We will show in the next analysis that the solution provided by the MoM simulator for different sizes of the larger patch in figure 4.8 approaching the infinite size is consistent with the image theory.

Three cases are investigated: the large patch, denoted *ground patch*(GP) has a size of $\lambda \times \lambda$, than $3\lambda \times 3\lambda$ and finally it is extended to an infinite ground plane. The $3\lambda \times 3\lambda$ maximum size was chosen as a compromise between simulation time and memory requirement. For this case the matrix filling time was approximative 6 min. and the inverting time was 25 sec. A number of 2000 basis functions were used. The excitation is a normal incident, plane wave, impinging on the patches from $+z$ direction with $E_x = 1 \frac{V}{m}$ in amplitude. Figure 4.10 shows cuts taken along the center of the small patch for the J_x surface current component magnitude value. The solution of the MoM simulator implemented here is in agreement with the theoretical solution given the infinite ground plane model through image theory. As we expect, the currents on the small patch suffer almost no influence by the fact that the patch below changes in size. The same results have been observed when a centered basis function was excited on the

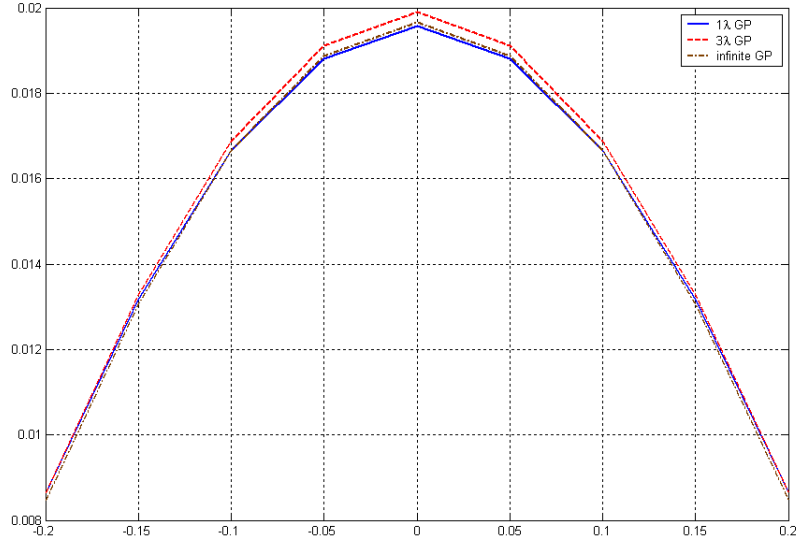


Figure 4.10: J_x surface current component for different ground patch sizes. $\phi = 0^\circ$ cut. x-polarized, normal incidence, plane wave.

small patch. For this case we chose to show the far field change as the ground patch grows in size. One expects the front to back ration to increase as the ground patch gets larger and that is seen in figure 4.11.

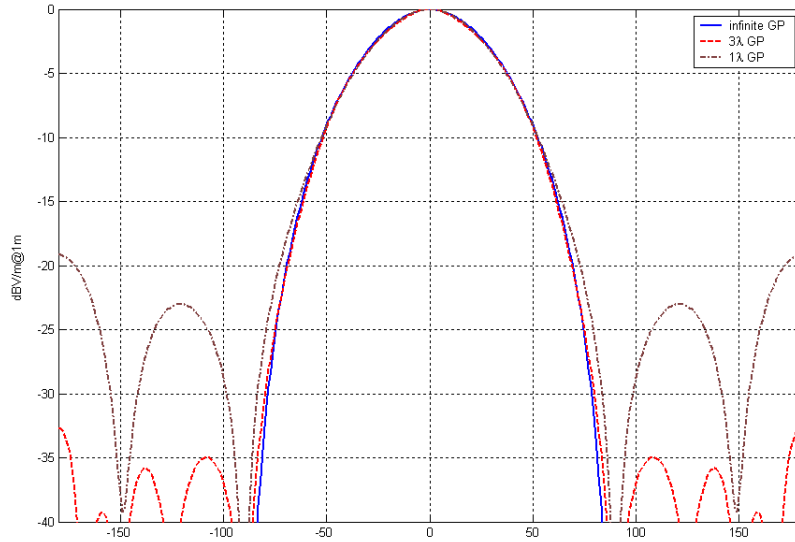


Figure 4.11: $E_{\theta}(\phi=0)$, scattered field for different ground patch sizes. centered x-directed basis function excitation on the small patch.

4.2 Verification with measurements

In this chapter we will show that there is a good agreement between the simulation results obtained with the moment method simulator and the real measurements. The probe excitation presented in the chapter (3) was elaborated in this sense and proves to be a reasonable model.

4.2.1 Two patch structure.

The goal was to measure a rather complex structure after optimization. The starting point was a two patch structure similar to the one in figure 3.4. The actual built antenna is presented in the next picture:

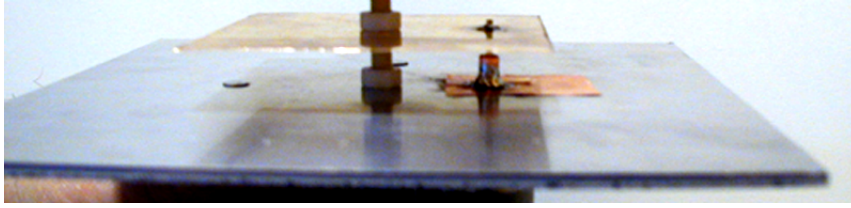


Figure 4.12: Two patch structure!

The physical parameters are:

-	size[λ^2]	size[mm]	spacing[λ]	spacing[mm]
ground patch	0.9×0.9	150×150	-	-
excited patch	0.4230×0.4230	70.5×70.5	0.0726	12.1

Table 4.4: Initial structure parameters. $\lambda = 166.66 \text{ mm}$ ($f_{rez} = 1814 \text{ MHz}$), see figure 4.14.

The probe was placed along the x -axis at a distance of approximately 11 mm from the edge and centered in y direction. This feed position is intended to give no cross-polarization in the E plane (xz plane in our case) and low cross-polarization in the H plane (yz plane).

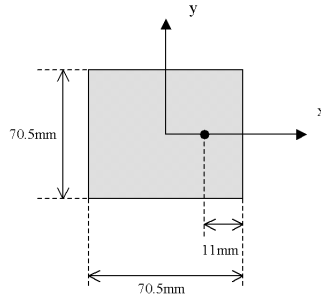


Figure 4.13: Probe position.

The resonance frequency for this structure was found experimentally by measuring the reflection coefficient S_{11} with the network analyzer. It has been found to be 1814 MHz as depicted in the following figure.

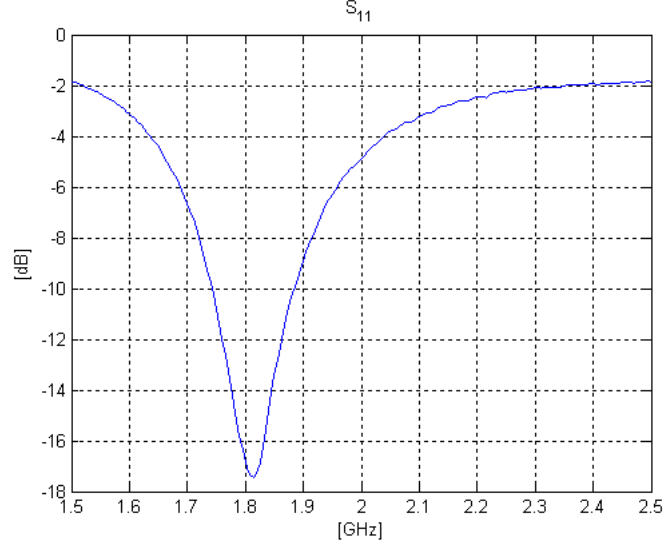


Figure 4.14: Measured reflection coefficient S_{11} for the two patch structure, figure 4.12.

-	$HPBW(\phi = 0^\circ)$	$HPBW(\phi = 90^\circ)$	$D_{max}[\text{dBi}]$	$F/B[\text{dB}]$
MoM	58.29°	67.2°	9.28	16.72
Meas.	54.5°	62°	$10.06^{\text{estimated}}$	16.9

Table 4.5: Radiation characteristics for the two patch structure, figure 4.12.

The radiation characteristics of this antenna are included in the table 4.5. The directivity in the case of the measured pattern was estimated using the data from the two cuts, E plane and H plane.

Figure 4.15 shows a comparison between the co-polar components of the E-plane, i.e. the E_θ components. As it can be seen the agreement with the measurements is very good in the forward radiation region and slightly worse in the back region. The front to back ration is also predicted correctly although one can notice some differences in the back lobe levels. This may come from the clamp fixing the antenna on the boom, signal socket or cable.

The MoM doesn't predict any cross-polar component for this cut, yet we see the measurement data has a cross-polar component. This is also due to the imperfections of the measurement set-up, radiation from the cables, etc...

Figure 4.16 shows the co-polar components of the H-plane, i.e. the E_ϕ components. A good agreement with measurements is also seen here with the same comments as above. The last figure shows the cross-polarization in the H plane. As one knows, cross-components are usually hard to predict. The

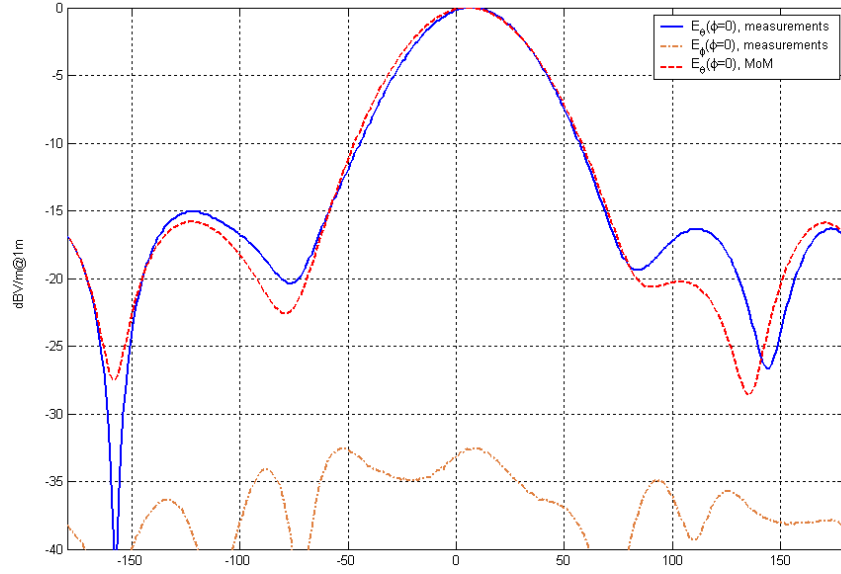


Figure 4.15: Comparison of measured and predicted E-plane co-polar and cross-polar far field components for the two patch structure presented in figure 4.12. Simulation mesh: 0.045λ ground patch, 0.0282λ excited patch.

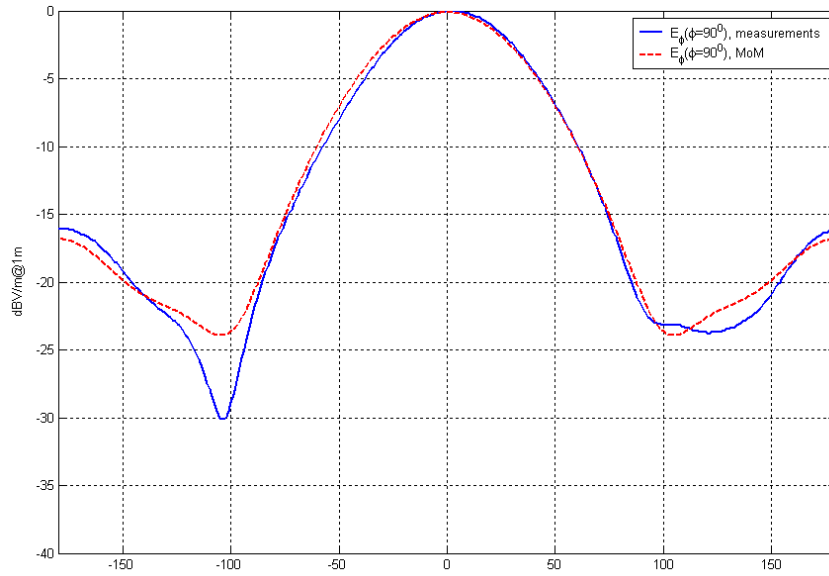


Figure 4.16: Comparison of measured and predicted H-plane co-polar far field components for the two patch structure presented in figure 4.12. Simulation mesh: 0.045λ ground patch, 0.0282λ excited patch.

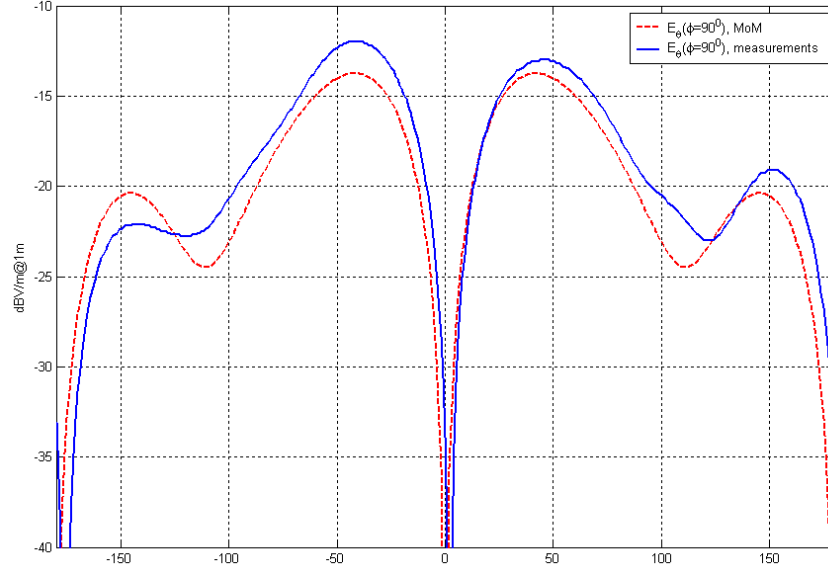


Figure 4.17: Comparison of measured and predicted H-plane cross-polar far field components for the two patch structure presented in figure 4.12. Simulation mesh: 0.045λ ground patch, 0.0282λ excited patch.

agreement in this case is quite good, although one can see the measurement data is slightly asymmetrical due to imperfect positioning.

4.2.2 Four patch structure.

Three more extra patches have been added to the previous structure. The patch size is fixed while their spacing were optimized for:

- Maximize directivity in a 20% frequency bandwidth around the resonance frequency.
- Maximize front to back ration. The target value was set to be 10dB.

The optimization procedure is described in detail in the following chapter. Only results will be given here. The probe position was unchanged. The optimized spacings are given below together with patch sizes.

Patch	length $[\lambda]$	spacing $[\lambda]$
excited patch	0.4230	0.0726
parasitic 1	0.38	0.08
parasitic 2	0.37	0.38
parasitic 3	0.36	0.05

Table 4.6: Patch spacings and sizes.

The actual structure is depicted in the following picture:

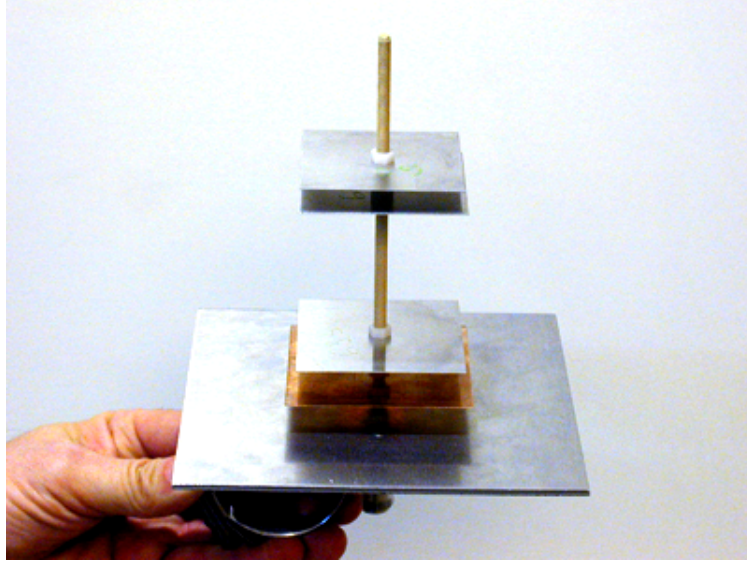


Figure 4.18: Four patch structure above finite ground reflector!

The reflection coefficient S_{11} in the $1.5GHz \div 2.5GHz$ frequency band is given in figure 4.19. As one can notice, the main resonance frequency has shifted to $1.625GHz$ and a second resonance is present at $1.995GHz$. The antenna ra-

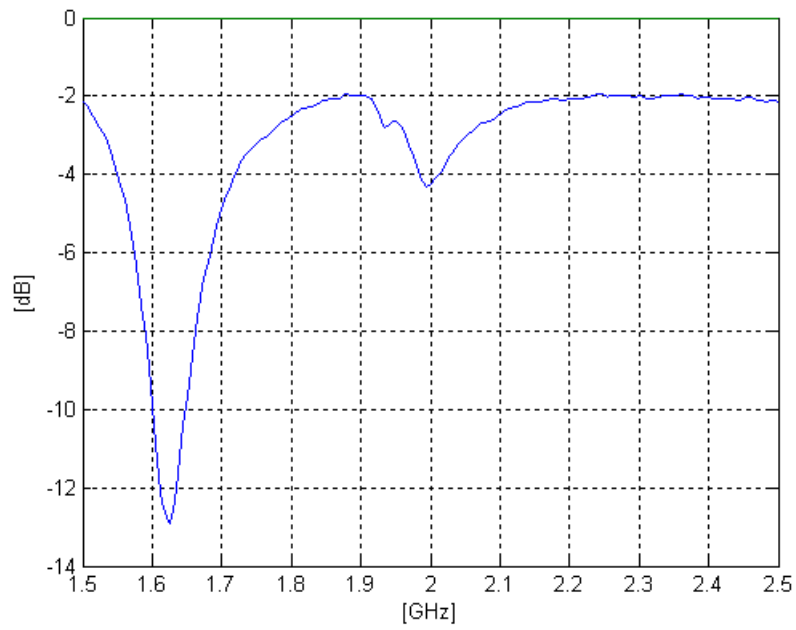


Figure 4.19: Reflection coefficient S_{11} for the 4 patch structure shown in figure 4.18.

diation pattern has been measured in the frequency interval: $1.7GHz \rightarrow 2GHz$ and results were compared with the ones predicted by the moment method simulator. The radiation characteristics at $1.73GHz$ of the optimized antenna are included in table 4.7. The directivity in the case of the measured pattern was

-	$HPBW(\phi = 0^0)$	$HPBW(\phi = 90^0)$	$D_{max}[dBi]$	$F/B[dB]$
MoM	48.7^0	50.92^0	10.04	9.56
Meas.	47.82^0	48.52^0	$10.49^{estimated}$	9.91

Table 4.7: Radiation characteristics at $1.7GHz$ for the 4 patch structure shown in figure 4.18.

estimated using the data from the two cuts, E plane and H plane.

We will present in the following figures a comparison between the measured and predicted patterns at $1730MHz$.

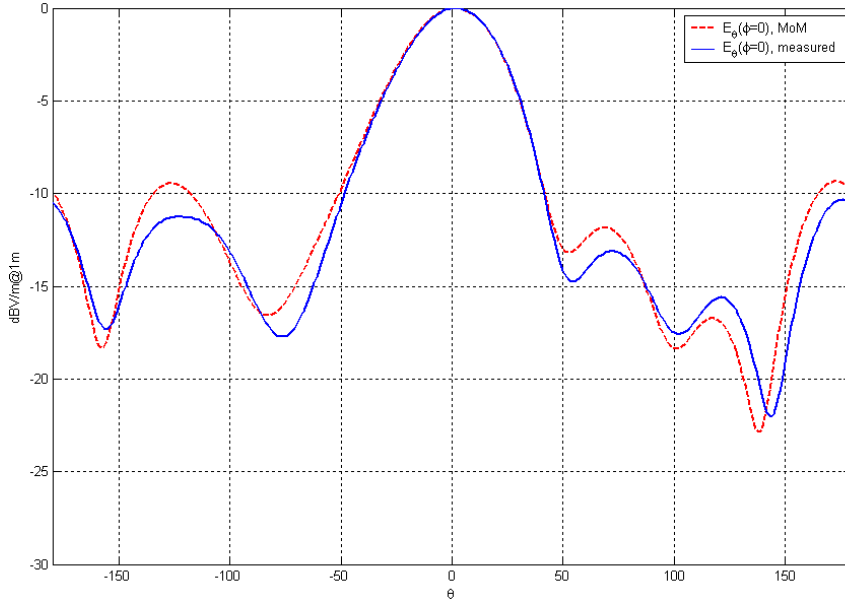


Figure 4.20: Comparison of measured and predicted E-plane co-polar far field components for the 4 patch structure shown in figure 4.18.

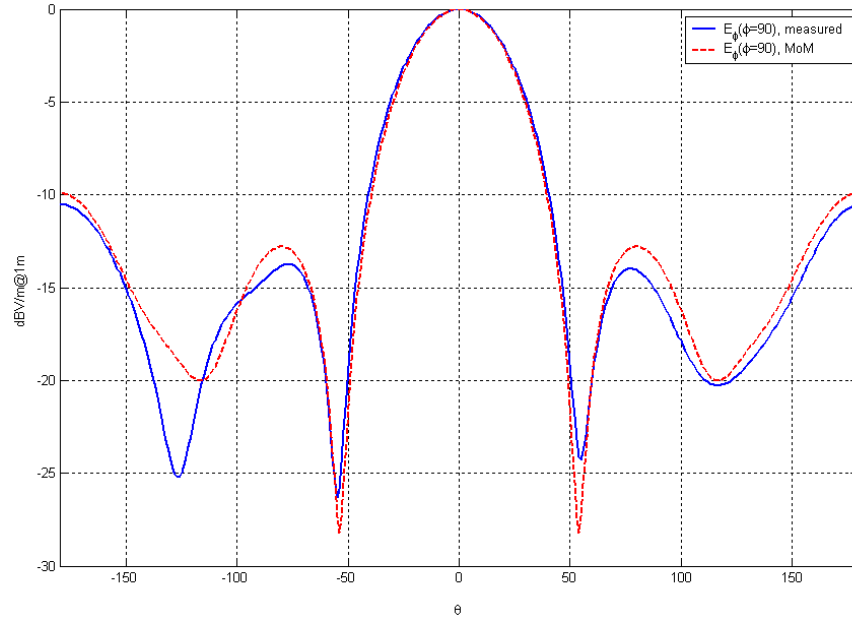


Figure 4.21: Comparison of measured and predicted H-plane co-polar far field components for the 4 patch structure shown in figure 4.18.

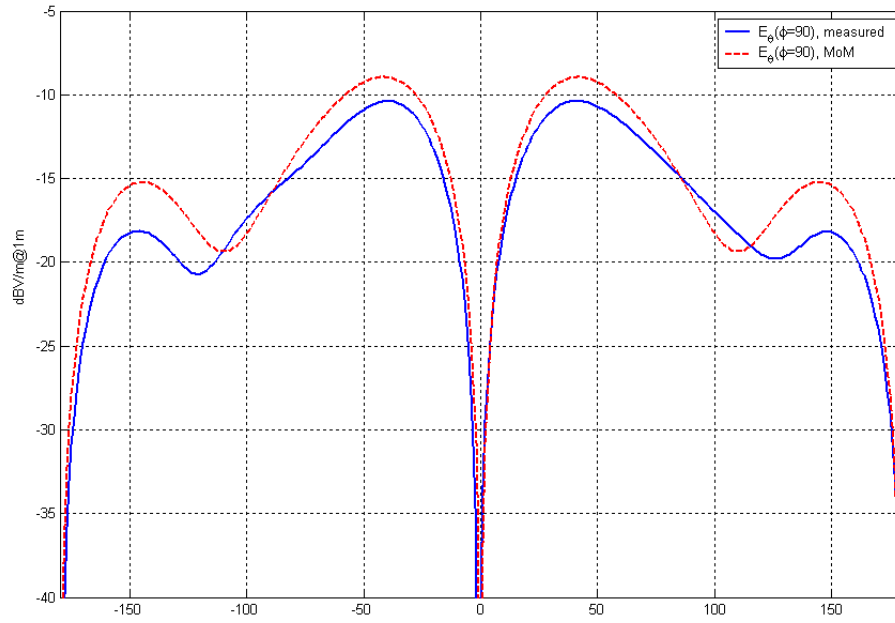


Figure 4.22: Comparison of measured and predicted H-plane cross-polar far field components for the 4 patch structure shown in figure 4.18.

Next we will show the frequency behavior of the measured $3dB$ beamwidth, estimated directivity and front to back ratio, while comparing them to the predicted values of the moment method simulator.

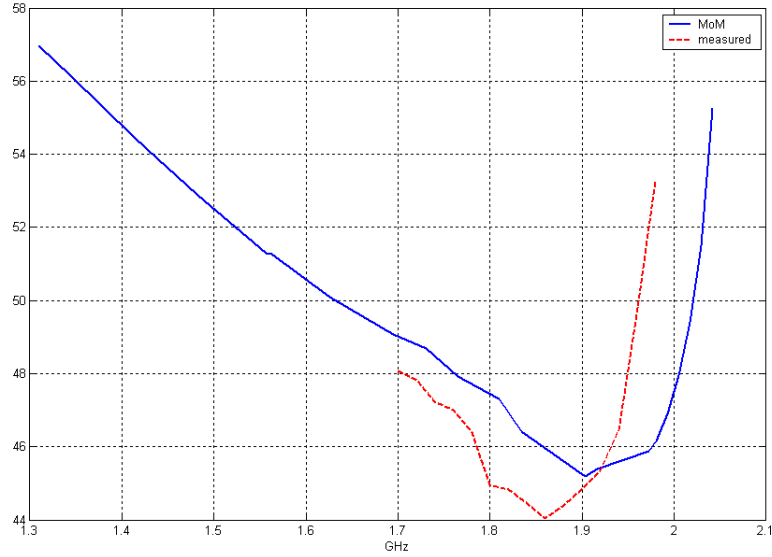


Figure 4.23: Comparison of measured and predicted E-plane half power beamwidth.

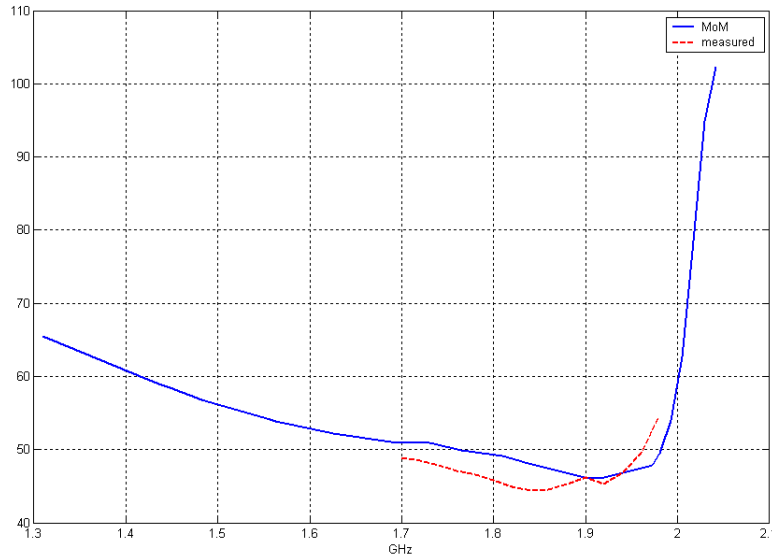


Figure 4.24: Comparison of measured and predicted H-plane half power beamwidth.

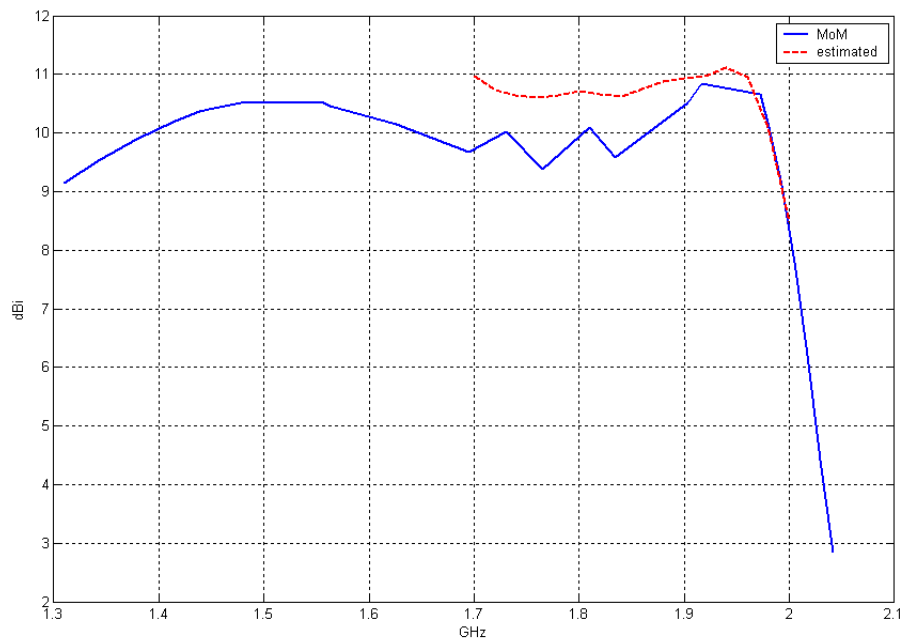


Figure 4.25: Comparison between directivity computed with MoM and estimated value from E - and H -plane far field measurements.

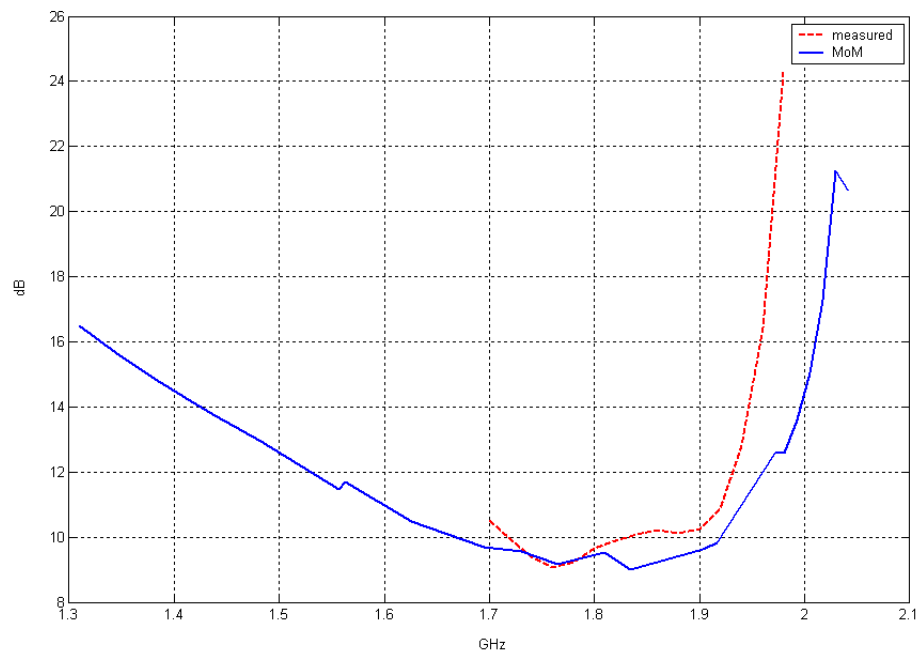


Figure 4.26: Comparison of measured and predicted front to back ratio.

Some words about the matching between patterns at other frequencies. There seems to be a better matching for frequencies closer to the resonance frequency. Also, close to the second resonance, the moment method simulator doesn't predict an end-fire radiation pattern, while in the measurements, it has large beamwidth in both cuts and low directivity. In conclusion we can say that the measured data is predicted by the simulator with good accuracy.

Chapter 5

Optimizations

Having presented the measurements results before, we can say that the moment method simulator predicts patterns that are in good agreement with the measurements, so one can use it as a tool to optimize these multi-patch structures. Since an antenna of this kind, having more than 3-4 parasitic patches is very difficult to optimize in practice, the MoM simulator will prove to be a valuable tool.

In this chapter we will try to give more insight on designing antennas with high directivity using the MoM simulator and draw conclusions about the actual coupling and radiation mechanism.

There are many guide lines to optimize antennas: maximum directivity at one frequency or in a frequency band, reduced side lobes, reduced cross-radiation, front to back ratio, etc... Usually a practical design will take into account all of these. In the previous chapter we tried to fulfill all of this requirements and find the best design with only four patches. In this chapter, we will concentrate only on optimizing for maximum directivity at one frequency.

We will use the assumption that the coupling mechanism doesn't depend on the feeding method. The patterns might be slightly different but the design will be optimal or close to optimal. For this reason, we will use a simpler excitation method than the probe, which will be presented in the following paragraphs.

One should also remember that usually a optimization process is quite slow, so for this matter we will profit of the good convergence of the solution given by this simulator and use a rather coarse mesh during optimization. When the optimal design has been found, we will show that using a finer mesh, results don't suffer major changes and solution converges fast.

There are two approaches to an optimal design which can be adopted:

- The first one considers a finite ground patch and tries to achieve maximum directivity. The risk in this case is that the radiation characteristics of this structure may be dependent of the reflector size, and changes in it's size may affect dramatically it's behavior.
- The second one considers the first patch(ground patch) to be infinite and tries to optimize the size and spacing of the other patch for highest directivity. In this way we can have a design which is independent of the size of the ground patch size, which is a good thing, and also can answer the

question: Does the multi-patch structure work with an infinite reflector?

5.1 Finite ground patch.

We have chosen a four patch structure. These four square patches are placed above a finite, fixed size ground patch and their spacing and sizes are the subject of optimization. For this we used a dedicated **Matlab** function, "fmincon.mat". The optimization process mainly consists in trying to minimize a certain constrain function for a given set of input parameters. In our case the input parameters are the patch sizes and spacings while the constrain function is directly related to the directivity of the structure so that it's minimization will give the highest directivity. One can see that this problem is in our case a 8 dimensional optimization problem which can be very time consuming and we want to avoid having a ill-posed problem. For this reasons, the first patch size and spacing above the ground patch are fixed and set to have values closed to real designs:

- ground size = $0.9\lambda \times 0.9\lambda$, z position 0
- excited size = $0.423\lambda \times 0.423\lambda$, z position = 0.0726λ

The excited patch is depicted in the following figure:

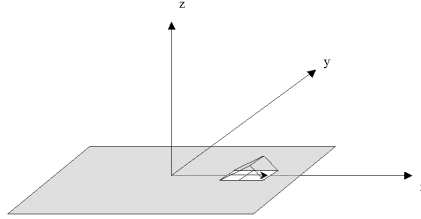


Figure 5.1: Patch with a x-directed basis function excitation.

A x -current basis function coefficient is set to $1[\text{A/m}]$. The basis function is centered in the y direction and placed approximately at $\frac{1}{3}$ distance from patch edge. This excitation is similar to the probe excitation, but much simpler and less time consuming.

The results of this optimization can be seen in table 5.1.

-	length[λ]	spacing[λ]
excited patch	0.4230	0.0726
parasitic 1	0.3534	0.4528
parasitic 2	0.3431	0.4136
parasitic 3	0.3739	0.3581

Table 5.1: Optimized sizes and spacings.

The radiation patterns for the E plane is presented in 5.2. The dashed

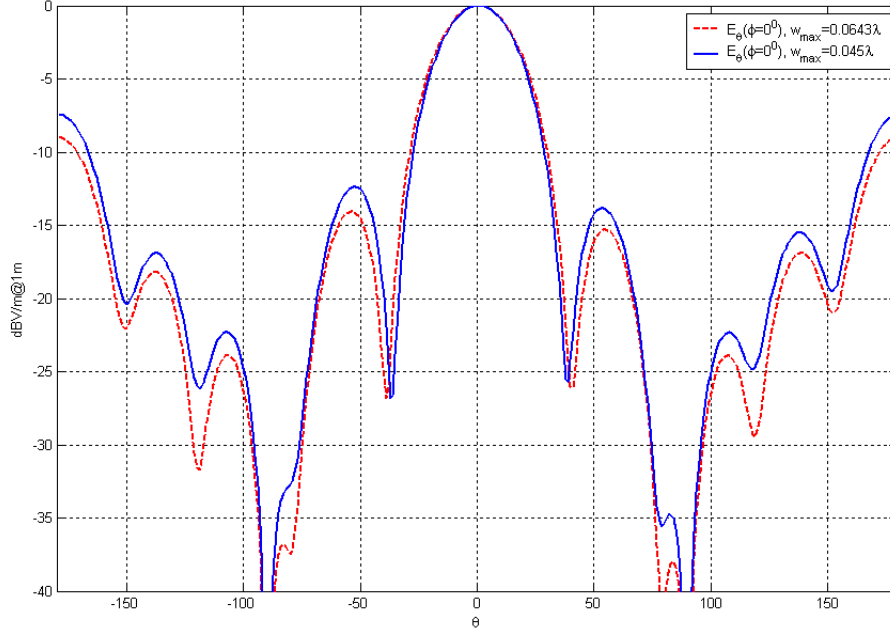


Figure 5.2: Solution convergence. Comparison of the E plane far field pattern for a optimization mesh of maximum cell width of $w_{max} = 0.0643\lambda$ and a finer mesh of minimum cell size of $w_{max} = 0.045\lambda$.

pattern in the previous figure has been used for optimization. The maximum mesh cell in this case is $0.0643\lambda \times 0.0643\lambda$. After finding the optimum design the mesh has been refined so that the cell size is less than $0.0450\lambda \times 0.0450\lambda$. This corresponds to the solid line. As anticipated before the changes in the pattern are not significant and optimization results converge to the same design even for a sparser mesh.

Figure 5.3 shows a comparison between the optimized pattern with a finite $0.9\lambda \times 0.9\lambda$, dashed line and an infinite ground patch, solid line. As one can see, the optimization for maximum directivity, led to a design with high back radiation, i.e. 7.42[dB] as stated in table 5.2. The replacement of the finite ground patch seems to introduce more side lobes and with higher levels, although the HPBW has narrowed.

The radiation characteristics of this structure are presented together with the one having a infinite ground patch in table: 5.2. Although we have optimized for a finite ground patch the parameters are still fine for the infinite ground patch.

In order to see how the patches interact with each other, we have defined equivalent dipole moments for each patch. The equivalent dipole moment for a patch is sought to represent the behavior of the patch currents and interaction with other patches. We compute the equivalent dipole moments as:

$$I_l(patch_j) = \sum_i J_x^i \cdot W \cdot L_{eq} \quad [\text{Am}] \quad (5.1)$$

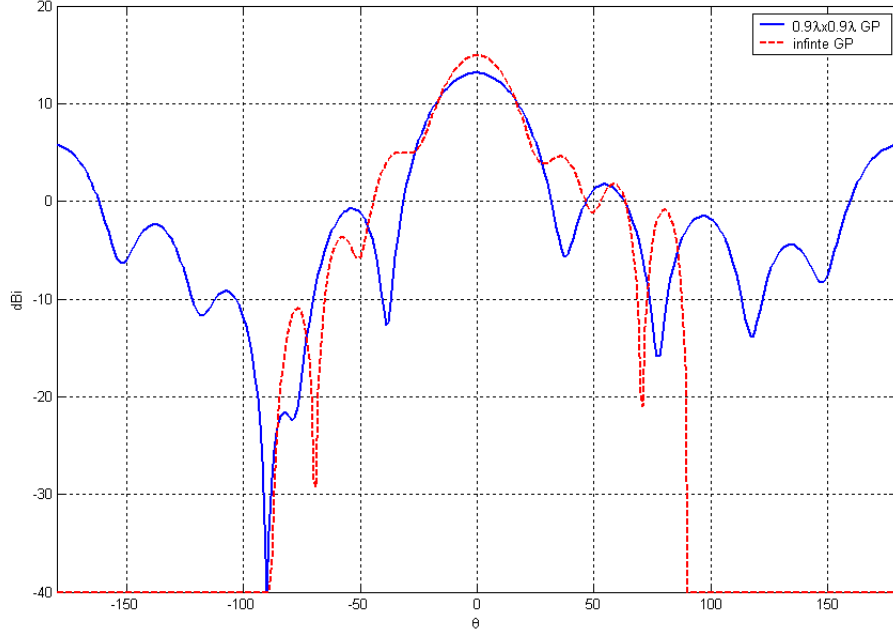


Figure 5.3: Directivity comparison between structure having a finite ground patch and infinite ground patch, $\phi = 0$ cut.

-	Infinite GP	finite GP
E plane, HPBW[deg]	26.6	33.95
H plane, HPBW[deg]	27.41	35.38
Maximum directivity[dBi]	14.94	13.12
Front/Back[dB]	Inf	7.42

Table 5.2: Radiation characteristics of the optimized multi-patch structure.

J_x^i [A/m] represents the surface current coefficient of the i^{th} current on $patch_j$, W [m] represents the current width (equal to the cell width) and L_{eq} [m] is the equivalent current length. The surface current spans as we know over two cells, but since we assume a triangular distribution, the equivalent current length is:

$$L_{eq} = \frac{2 \times \text{cell length}}{2} \quad (5.2)$$

The J_y currents for this kind of excitation are symmetrical and their equivalent moment will be 0.

The following tables present the dipole moments as defined above, with amplitudes normalized to the current amplitude of the excited patch.

One can see in the table above that the optimization led, like in the example with the Yagi-Uda array from [2], to a design where current amplitudes decrease monotonically from the excited element as well as the phase. This indicates that the structure is supporting a travelling wave with a phase velocity that decreases as we move away from the excited element. The ground patch has the effect of

patch	Infinite ground patch		Finite ground patch	
	Dip. Mom.[Am]	$\Delta\phi$	Dip. Mom.[Am]	$\Delta\phi$
ground	-	-	$1.2286\angle -174.9^\circ$	-
excited	$1\angle 7.177^\circ$	-	$1\angle 9.6^\circ$	175.4°
parasitic 1	$0.346\angle -173.1^\circ$	180.3°	$0.322\angle -170.6^\circ$	180.2°
parasitic 2	$0.237\angle 15^\circ$	171.8°	$0.215\angle 15.7^\circ$	173.6°
parasitic 3	$0.171\angle -146.8^\circ$	161.8°	$0.155\angle -147.2^\circ$	163°

Table 5.3: Normalized equivalent dipole moments. $\Delta\phi = \phi_i - \phi_{i+1}$.

a reflector for the travelling wave and the shift to an infinite reflector doesn't change very much the current distribution on the patches above. In this way we can explain the fact that the radiated pattern still resembles the one from the finite case. Also, one can notice that for the infinite case, the phase decreases more rapidly, i.e. the phase delay is larger, the effect being a pattern with higher directivity. If we consider the array factor for an array of elements along z-axis:

$$AF(\theta) = \sum_{i=1}^N I_i \cdot e^{j(kd_i \cos \theta + \beta_i)} \quad (5.3)$$

we can construct a phase diagram of the currents considering their amplitudes and phase coefficients β_i . The following figure is the phase diagram considering only the currents on the excited element and parasitic elements. The currents of the reflector and image patches were disregarded since the main purpose of the figure is to show that an increased delay leads to more directivity.

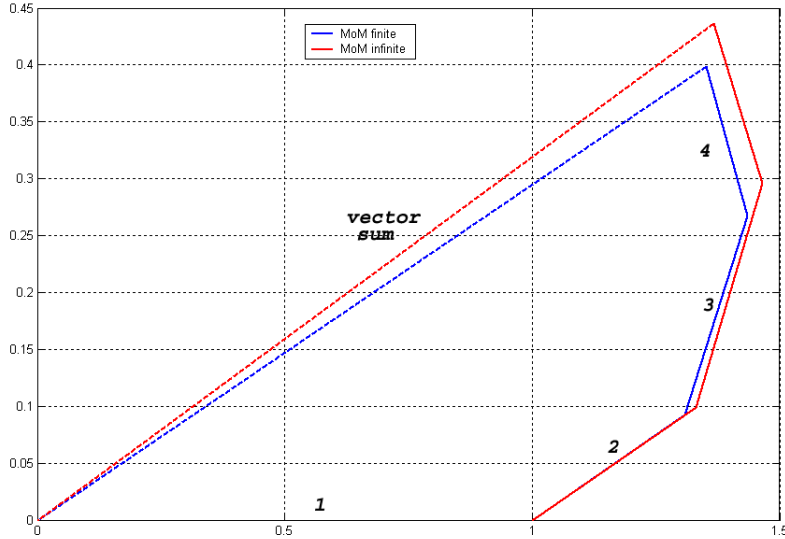


Figure 5.4: Phase diagram of normalized equivalent dipole moments for the active patches in the infinite and finite ground patch cases. Vector phases are the dipole's phase coefficients β_i .

5.1.1 Hansen-Woodyard array comparison.

A comparison between the optimized array and the Hansen-Woodyard array follows in this section.

The Hansen-Woodyard condition for uniformly spaced and very long, arrays, states that to achieve a higher directivity than the ordinary end-fire array, the phase shift of element i should be :

$$\beta_i = -(i-1)(kd + \frac{2.94}{N}) = -(kd_i + (i-1)\frac{2.94}{N}) \quad \text{for } \theta = 0^\circ \quad (5.4)$$

N represents the number of elements and d the spacing. The above condition will increase the directivity of the ordinary end-fire array with:

$$10 \log_{10}(1.789) = 2.526[\text{dB}]$$

This is a super-directivity condition, but, the Hansen-Woodyard condition doesn't necessary yield the maximum directivity and the best front to back ratio. It is just a condition to get more directivity than the normal end-fire condition.

We will compare the directivity of our antenna for the infinite ground patch case with the directivity of an ordinary uniform amplitude and spacing end-fire array and the same array with progressive phases from the Hansen-Woodyard condition. For this, we have to consider an equivalent array of 8 elements, i.e. to include the image patches. From the beginning we should mention that in our case we can excite only one element, while for the uniform arrays we can control the amplitude and phase of each current element. We should anticipate that these arrays should behave better than ours for this reason. We considered the amplitude of each element of the uniform array to be 1[A].

The following table shows the three arrays with the current amplitude and phases coefficients β_i of each element and also the elements z-position:

MoM		
	Amp(β_i)	z-pos[λ]
el. 1	-0.1713(0°)	-1.297
el. 2	-0.2375(32.94°)	-0.939
el. 3	-0.3462(55.89°)	-0.525
el. 4	-1(73.2°)	-0.072
el. 5	1(20.93°)	0.072
el. 6	0.3462(37.6°)	0.525
el. 7	0.2375(76.8°)	0.939
el. 8	0.1713(146.08°)	1.297

Table 5.4: Normalized equivalent dipole moments of the optimized array considering also the image patches.

Using the above data we have computed the directivity of each equivalent array considering its elements to be isotropic. The results are plotted in figure 5.5. We can see that the optimized array performance is closed to the ordinary end-fire array, but worse. As expected, the Hansen-Woodyard array performs better than our array since all current amplitudes are equal and phases are according to equation 5.4. This comparison is useful in providing a theoretical limit to the

	Ordinary		Hansen-Woodyard	
	Amp(β_i)	z-pos[λ]	Amp(β_i)	z-pos[λ]
el. 1	1(0°)	-1.2970	1(0°)	-1.2970
el. 2	1(133.40°)	-0.9264	1(154.46°)	-0.9264
el. 3	1(-93.19°)	-0.5559	1(-51.07°)	-0.5559
el. 4	1(40.21°)	-0.1853	1(103.38°)	-0.1853
el. 5	1(173.62°)	0.1853	1(-102.16°)	0.1853
el. 6	1(-52.97°)	0.5559	1(52.3°)	0.5559
el. 7	1(80.43°)	0.9264	1(-153.23°)	0.9264
el. 8	1(-146.16°)	1.2970	1(1.23°)	1.2970

Table 5.5: Currents amplitude and phase coefficients β_i for uniformly spaced arrays with ordinary and Hansen-Woodyard end-fire condition.

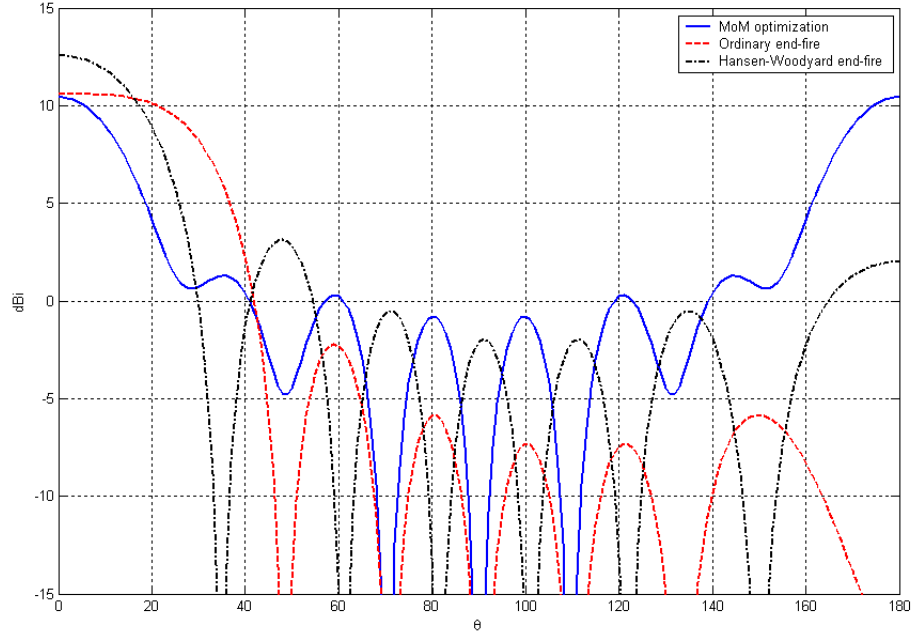


Figure 5.5: Directivity comparison between arrays of uniformly spaced, constant amplitude, isotropic elements with ordinary and Hansen-Woodyard end-fire condition and an equivalent array of isotropic elements with currents values and positions defined in table 5.4.

directivity of our array. It should be mentioned that to achieve this, we have to control the phase and amplitude of currents on each patch, but even so there is a restriction imposed by the image theory for the image currents phase shift.

5.2 Infinite ground patch.

Using the same excitation and constrain function we now start the optimization process considering the ground patch to be infinite. Results are given as before in comparison with a finite ground patch of $0.9\lambda \times 0.9\lambda$. The optimized sizes and spacings can be seen in the next table:

-	length[λ]	spacing[λ]
excited patch	0.4230	0.0726
parasitic 1	0.3895	0.2861
parasitic 2	0.4308	0.0868
parasitic 3	0.4548	0.4303

Table 5.6: Optimized sizes and spacings values for infinite ground patch.

The radiation characteristics are given in the next table:

-	Infinite GP	finite GP
E plane, HPBW[deg]	28.54	34.6
H plane, HPBW[deg]	29.89	34.54
Maximum directivity[dBi]	14.77	10.64
Front/Back[dB]	Inf	3.84

Table 5.7: Radiation characteristics comparison.

The normalized equivalent dipole moments are given next:

patch	Infinite ground patch		Finite ground patch	
	Dip. Mom.[Am]	$\Delta\phi$	Dip. Mom.[Am]	$\Delta\phi$
ground	-	-	$1.532\angle -176.04^\circ$	-
excited	$1\angle 9.67^\circ$	-	$1\angle 12.14^\circ$	171.8°
parasitic 1	$0.853\angle -0.67^\circ$	10.35°	$0.78\angle -1.03^\circ$	13.18°
parasitic 2	$1.247\angle -179.33^\circ$	178.66°	$1.132\angle -177.95^\circ$	176.91°
parasitic 3	$0.3542\angle 1.04^\circ$	179.64°	$0.309\angle 0.53^\circ$	181.51°

Table 5.8: Normalized equivalent dipole moments. $\Delta\phi = \phi_i - \phi_{i+1}$

Same conclusions can be drawn when comparing the above results. Also one notices that the optimal design in this case performs worse than the initial design in terms of front to back ration, beamwidth and directivity. The optimization process was also very long and it seems that there are many designs which get close to the radiation characteristics from table 5.7.

5.3 Ground patch size influence

As we said in the beginning we want to see how sensitive is the optimal design to the ground patch size. We choose the design from the finite ground plane optimization. The patches parameters are kept constant while the ground patch length is changed from 0.5λ to 2λ .

Figure 5.6 shows the directivity and front to back ration variation with the ground patch size. As we expect the front to back ration is very dependent on the ground patch size. On the other hand the directivity seems to vary less, i.e. from 0.6λ to 2λ there was a gain of 2dB only in directivity. Also the gain to the directivity in the case of infinite ground patch size is 3dB.

The HPBW for both cuts is, as expected, quite dependent on the ground patch size, so from this point of view the design is sensible to the ground patch size.

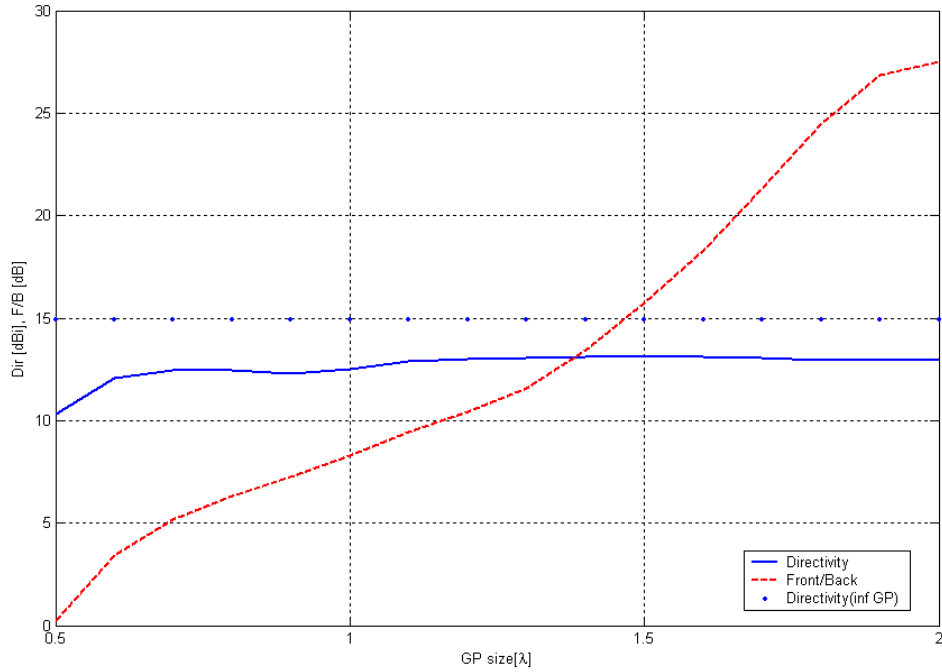


Figure 5.6: Directivity and Front to back ration dependence on the ground patch size.

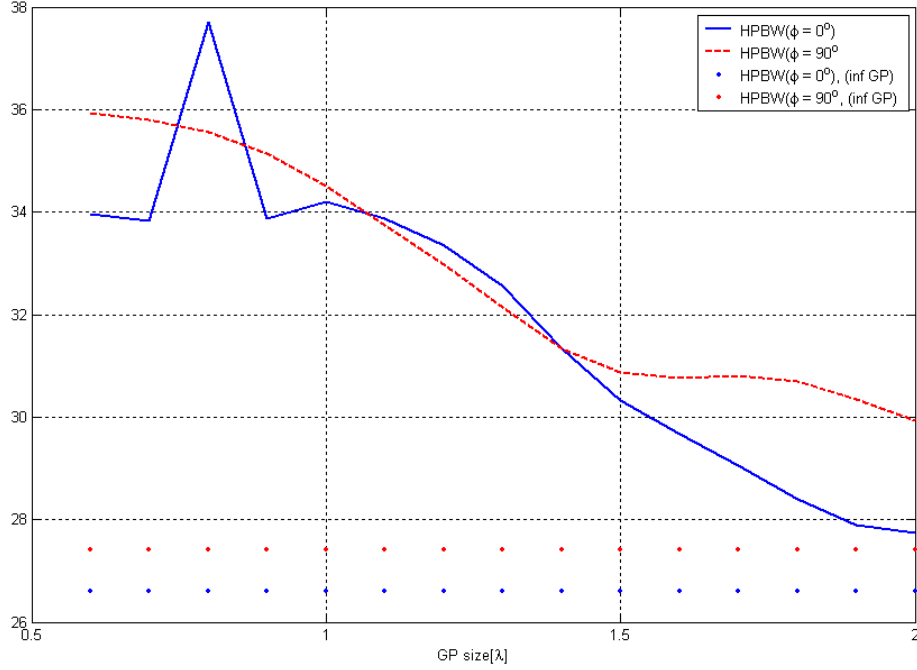


Figure 5.7: HPBW for $\phi = 0$ and $\phi = 90$ dependence on the ground patch size.

We can see that using the approach of infinite ground plane optimization is not very good since designs seem to perform bad when finite ground patch is added, but we notice that the radiation mechanism is mainly the same in both cases and an infinite or a large ground patch can increase directivity with up to 2-3dB.

The multi-patch structure works also with infinite ground patch. The most sensitive parameters to the ground patch size are the front to back ratio and the HPBW. Directivity on the other hand may vary less for a wide range of ground patch sizes.

We also saw that the conclusions drawn from the Yagi-Uda arrays can be applied in this case too and the multi-patch patch structure can also be regarded as a structure supporting a travelling wave. We had also showed that there is a relation between the phase delay between elements and the radiation mechanism, i.e. increasing the phase delay, directivity increases.

The equivalent current moments are a useful tool in studying this structure providing insight regarding the coupling between patches and also the radiation pattern. It can be also useful to compare results with known results of linear arrays.

Chapter 6

Conclusions and future work

In this thesis work we have developed a theoretical model of a multi-patch structure and tried to provide insight on how patches couple together and contribute to the radiation pattern. A full-wave approach was used. A system of integral equations was set up and solved using the moment method. Plane wave and probe feed excitation were implemented. The predicted results for the plane wave excitation were verified against literature and for the probe feed model measurements were done. Predicted results show very good agreement with measurements.

The initial requirements on high directivity over a broad frequency band can be met using the multi-patch antenna. We have achieved:

- directivity of 14.94dBi
- E – and H –plane HPBW of approximately 27°

for a 4 patch structure having an infinite reflector and:

- directivity of 13.12dBi
- E – and H –plane HPBW of 34° and 35.4°
- front to back ration of 7.42dB

for a 4 patch structure having a $0.9\lambda \times 0.9\lambda$ reflector.

Our results lead to the conclusion that the Yagi-Uda studies presented in the literature provide valuable hints on how to approach the study of multi-patch structures. Although the above investigations cannot thoroughly prove so, they suggests that these type of structures might support slow travelling waves. Further investigation are needed.

The implementation of aperture feed can provide valuable results also, as in practice it is more often preferred than the probe feed excitation.

Also a proper study on dual polarization feeding options is required. Only structures providing a single polarization field were investigated. A dual polarized structure can be easily built starting from the previous models by having a second excitation feed for the orthogonal plane. The aperture feed is expected to perform better from this point of view.

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Appendix A

A.1 Numerical comparison

We present in the following section the first column of the Z matrix, values that were calculated for a 5×2 cells structure as depicted in figure A.1. The cell size is $0.1\lambda \times 0.1\lambda$. Comparison is made between data provided by Andrew F. Peterson and the MoM simulator.

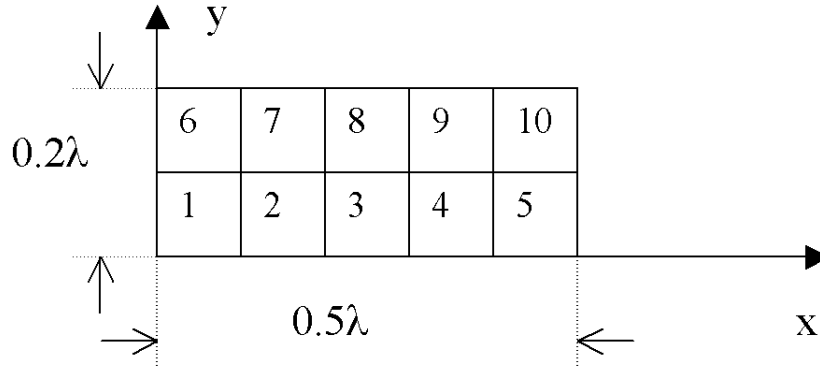


Figure A.1: J_y surface current component, real part, for a $\lambda \times \lambda$ patch in free space. x-polarized, normal incidence, plane wave.

The difference noted between the two sets of values is due to the different approaches. Peterson's method, presented in [14], p.420 uses a formalism involving only vector potentials. Reaction integrals involving Green's function are regarded like two dimensional convolution, and differentiations are carried out analytically. This approach yields more accurate results but is not suited for us, from the point of the convolution process when the domain is extended to three dimensions.

	<i>Peterson</i>	<i>MoM*</i>
$Z_{1,1}$	$7.761658 - 196.4987i$	$7.64043 - 206.90357i$
$Z_{2,1}$	$7.462346 + 113.1230i$	$7.35533 + 117.31015i$
$Z_{3,1}$	$6.613650 + 23.73244i$	$6.54623 + 24.37035i$
$Z_{4,1}$	$5.352869 + 5.973901i$	$5.33856 + 6.225246i$
$Z_{5,1}$	$7.162387 - 21.94106i$	$7.04610 - 23.23374i$
$Z_{6,1}$	$6.879745 + 8.734553i$	$6.77739 + 8.778979i$
$Z_{7,1}$	$6.078915 + 10.34532i$	$6.01451 + 10.63566i$
$Z_{8,1}$	$4.891143 + 3.741937i$	$4.87827 + 3.93742i$
$Z_{9,1}$	$0.074641004 + 106.3816i$	$0.07453 + 106.46296i$
$Z_{10,1}$	$-0.074641004 - 106.3816i$	$-0.07453 - 106.46296i$
$Z_{11,1}$	$-0.2115536 - 14.26167i$	$-0.21169 - 14.26759i$
$Z_{12,1}$	$-0.3139639 - 2.686226i$	$-0.31418 - 2.68817i$
$Z_{13,1}$	$-0.3670754 - 0.9713796i$	$-0.367331 - 0.97204i$

Table A.1: First column of the Z matrix. Comparison between MoM results and values provided by Andrew F. Peterson, [18].*MoM results have been scaled with the cell width(0.1λ) to have a fair comparison with Peterson results.

Appendix B

B.1 Simulator's graphical interface

A simple graphical interface was developed using **Matlab** gui functions. It's main purpose is to facilitate access to results for a fluent and efficient analysis. There are some restrictions when building the structure that need to be stated. Always have plate length in x direction less or equal the plate length in y direction. This is used in the $\mathbf{Z}$ matrix filling to speed up things. The minimum number of mesh units in one direction is 3 from plotting considerations.

The first step in the simulation is to fill in the input parameters in the "main.m" file. These are:

- Number of patches: "in.patch_nr".
- Number of cells in x and y direction for each patch: "in.X_len" and "Y_len".
- Z coordinate for each patch, normalize value with λ : "in.Z".
- Cell size for each patch, normalize value with λ : "in.Cell".
- The infinite ground-plane option: "in.gp". "0" if no ground plane is considered, and "1" if infinite ground plane is considered. By default, the $z=0$ plane is considered the GP, so one should be careful when entering the z positions of the patches in this case.

Running the "main.m" script will start the computation of the $\mathbf{Z}$ matrix. The **Matlab** prompt window will show the filling time for the **A**, **B**, **C** and **D** matrixes separately. Finally, the inverting time is presented. At this moment the $\mathbf{Z}$ matrix is computed and inverted so different excitations may be chosen to compute results. For this the graphical interface must be started. This is done by running the "gui.m" script file.

We enumerate the excitation options now:

- Plane wave excitation. It considers as mentioned in the previous chapters a normal incident, plane wave, coming from $+z$ direction. The E_x and/or E_y components amplitudes can be chosen.
- Probe excitation. For the probe excitation, we have to choose the cell which contains the feeding point by filling in the "x_position" and "y_position". The feed point is considered to be centered in this cell.

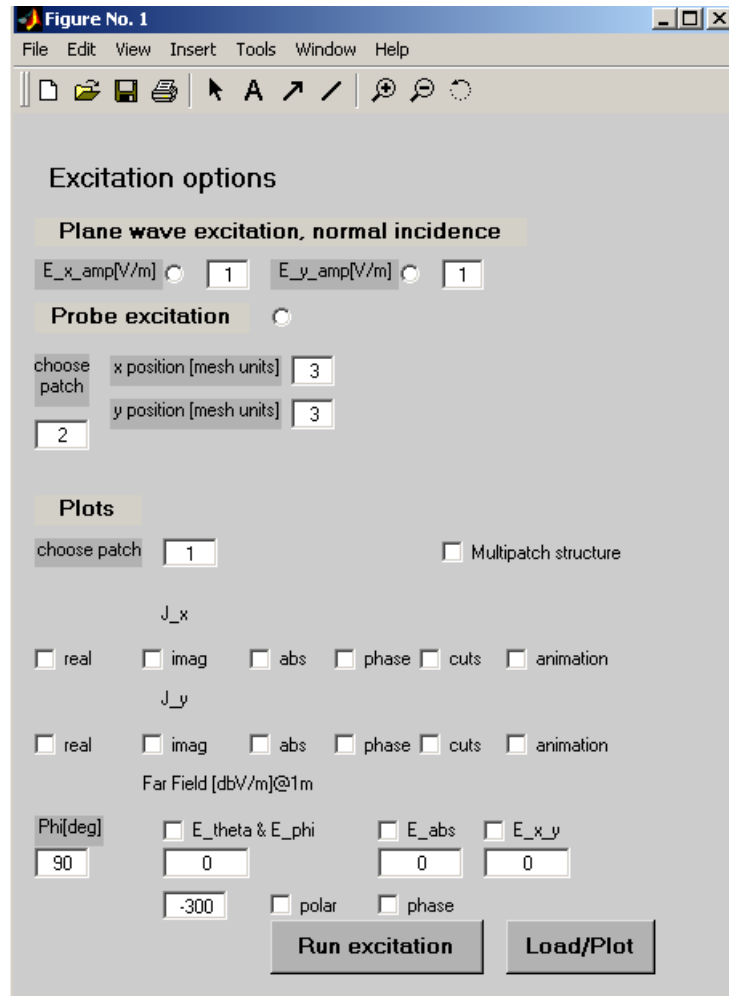


Figure B.1: Graphical interface for the MoM simulator.

The **Run Excitation** button considers the above selected excitation and calculates the $\mathbf{V}$ vector and then the surface equivalent current coefficients.

To view the results the **Load/Plot** button must be used. One can choose to see plots of surface currents. Separately real part, imaginary part, magnitude, phase or centered cuts are available for these currents. The far field can be viewed for different ϕ cuts. Spherical and rectangular components are available. For the plane wave excitation only the scattered field is plotted since there is no definition of total field in this case. The plane wave field is constant everywhere, while the scattered field is attenuated with distance.